

Empirical Finance Porfolio Project

Camila Janada

Professor Peterson

ECON 4360

May 8, 2026

Contents

Exercise Set 1	3
Problem 1.1	3
Problem 1.2	4
Problem 1.3	5
Problem 1.4	5
Conclusion	6
Exercise Set 2	7
Exercise Set 3	9
Exercise Set 4	17
Question 4.1	17
Question 4.2	19
Exercise Set 5	21
Exercise Set 6	25
Portfolio	25
Question 1: RiskRets and PortRisk	25
Question 2: AQR Factors	26
Question 3: Efficient Frontier	28
Question 4: Tech Bubble of the late 1990's?	29
Question 5: Accumulated Wealth	30
(a) Accumulated wealth	30
(b) IQR of Terminal Wealth	30
(c) 5% VaR	31
Exercise Set 7	32
Question 1: Terminal Distribution of Wealth	32
Question 2: Interquartile Range	32
Question 3: What is the risk (likelihood) of having less than \$P in terminal wealth? . . .	33
Question 4-5: What would a retirement annuity look like in its simplest terms (in dollars per year, say)?	33
Question 6: Suppose there is a chance during retirement that our retiree will be hit with an unexpected medical expense of \$M that could occur in any year. Could we somehow account for the impact of that?	34
Exercise Set 8	36
Question 8.1:	36
Question 8.2:	37
a. Price a ten percent OTM call	37
b. Price a ten percent OTM put	38
c. Compute the delta for the call	38
d. Compute the delta for the put.	38
Question 8.3:	39
Question 8.4:	39

Question 8.5:	39
Question 8.6:	40
Question 8.7:	40
Question 8.8:	41
Exercise Set 9	42
Section 1:	42
Section 1 Conclusion	42
Section 2:	42
Section 2 Conclusion	43
Section 3:	43
Section 3 Conclusion	43
Section 4:	43
Section 4 Conclusion	44
Section 5:	44
Section 5 Conclusion	45
Conclusion:	45
Exercise Set 10	47
Appendix	55
Exercise Set 3: MatlabOLStutorial_26.m	55
Exercise Set 5.m	61
Exercise Set 6: OptPort_completed-1.m	68
Exercise Set 7.m	70
Exercise Set 9: portInsuranceHedge.m	73
Exercise Set 10: deltaHedgeTemplate.m	76

Exercise Set 1

First, go to Portfolio Outline and be sure to complete parts 1-7. When you've finished these, open Exercise Set 1 (2026).xlsx and complete Prob 1, Prob 2, Prob 3, and Prob 4 in the spaces provided in the spreadsheet. Upload your completed spreadsheet titled Exercise Set I (2026).xlsx to Canvas.

Problem 1.1

Table 1: Performance Metrics by Asset Class

	MSCI ACWI	GOLD	CBOE PUT WRITE	US AGG	US Corp HY	US TREAS	EW Portfolio
Geometric Mean	0.004775	0.005434	0.007013	0.003678	0.005636	0.003373	0.006121
Geo Mean Quarterly	0.014393	0.016390	0.021186	0.011075	0.017005	0.010155	0.018475
Mean Quarterly Ret Ann.	0.017319	0.019292	0.022443	0.011249	0.017838	0.010390	0.019685
Geo Mean Annualized	0.058828	0.067190	0.087476	0.045043	0.069775	0.041241	0.075974
Stdev	0.153564	0.155668	0.104538	0.041442	0.085591	0.046541	0.102213
Sharpe	0.383085	0.431625	0.836783	1.086899	0.815211	0.886129	0.743288

Problem 1.2

A major lottery pays the winner \$10 million. However, the prize money is paid out at the rate of \$500,000 each year with the first payment immediately for a total of 20 payments. What is the present value of this prize at 10% interest?

Table 2: Present Value of Cash Flows

t	d	CF	PV
0	1.00000000	500000	500000.00
1	0.90909091	500000	454545.45
2	0.82644628	500000	413223.14
3	0.75131480	500000	375657.40
4	0.68301346	500000	341506.73
5	0.62092132	500000	310460.66
6	0.56447393	500000	282236.97
7	0.51315812	500000	256579.06
8	0.46650738	500000	233253.69
9	0.42409762	500000	212048.81
10	0.38554329	500000	192771.64
11	0.35049390	500000	175246.95
12	0.31863082	500000	159315.41
13	0.28966438	500000	144832.19
14	0.26333125	500000	131665.63
15	0.23939205	500000	119696.02
16	0.21762914	500000	108814.57
17	0.19784467	500000	98922.33
18	0.17985879	500000	89929.39
19	0.16350799	500000	81753.99
		Total PV	\$4,682,460.05

Problem 1.3

You are one of a group of potential investors contemplating a ten-year lease of an oil well. The well produces 10,000 barrels of oil annually at a cost \$30 per barrel. The current spot price of crude is \$75 per barrel. The opportunity cost of capital is an annual 4%.

a) What is the value of the lease?

b) Suppose these investors have \$3 million to buy the lease. What is the lease's NPV?

Table 3: Discounted Profit Cash Flows

Year	Discount Rate	Profit	PV
1	0.961538462	450000	432692.31
2	0.924556213	450000	416050.30
3	0.888996359	450000	400048.36
4	0.854804191	450000	384661.89
5	0.821927107	450000	369867.20
6	0.790314526	450000	355641.54
7	0.759917813	450000	341963.02
8	0.730690205	450000	328810.59
9	0.702586736	450000	316164.03
10	0.675564169	450000	304003.88
		Value	\$3,649,903.10
		Net PV	\$649,903.10

Problem 1.4

Gross returns are denoted by

$$R = \frac{X_1}{X_0}$$

where X_1 is the total value of an asset at the end of period 1; X_0 denotes period 0 where the time interval from period 0 to period 1 is an arbitrary unit of time (a day, a month, etc.). Suppose the gross return is equal to 1.05 and the time interval is one year. The net return,

$$r = 0.05.$$

Now compute the effective returns if interest is paid semi-annually, quarterly, monthly, and continuously.

Periods/Year	Net Return
2	0.050625
4	0.0509453
12	0.0511619
$e^r - 1$	0.0512711

Conclusion

Problem 1 reveals that risk-adjusted performance, not raw return, is the appropriate ranking criterion, as it earns the highest Sharpe ratio (0.837) despite only a mid-range annualized return of 8.7%. Problems 2 and 3 demonstrate the power of discounting. A nominally \$10 million lottery prize decreases to \$4.68 million in present value, while the oil well lease generates a positive NPV of \$649,903, confirming that an asset is worth acquiring whenever discounted cash flows exceed its cost. Problem 4 completes the picture by showing that compounding frequency matters even at a fixed stated rate, with continuous compounding yielding an effective return of 5.127% against the semi-annual figure of 5.063%.

Exercise Set 2

This is a relatively short exercise set that will teach you to "rolling window" trailing returns plus using the current spot rates to "imply" a forward curve and forward rates. The Term Structure at any point in time "implies" a set of forward rates. The forward rates are the market's current priced-in forecasts of future rates at each maturity. Part 1: Estimate annualized returns for each index. Estimate the cumulative returns in percent for the three series. Estimate the annualized return standard deviations for the three indexes. Part 2: Linearly interpolate spot the yields for the maturities. Solve for the short forward rate.

$$\text{Rolling Return} = \exp \left(\sum_{i=2}^n \ln \left(1 + \frac{B_i}{100} \right) \right) - 1$$

$$\text{Cumulative Return} = \exp \left(\sum_{i=2}^{25} \ln \left(1 + \frac{B_i}{100} \right) \right) - 1$$

$$\text{Annualized Return} = \left(\exp \left(\sum_{i=2}^{25} \ln \left(1 + \frac{B_i}{100} \right) \right) \right)^{12/24} - 1$$

$$\text{Annualized Std. Dev.} = \text{STDEV}(B_2 : B_{25}) \times \sqrt{12}$$

$$\frac{(\text{Year}_2 \times \text{Rate}_2) - (\text{Year}_1 \times \text{Rate}_1)}{\text{Year}_2 - \text{Year}_1}$$

Date	MSCI US MOM	MSCI US VALUE	MSCI US MIN VOL			
Jan-16	-3.69	-4.35	-1.39			
Feb-16	-1.31	-0.19	1.09			
Mar-16	5.28	6.68	6.01			
Apr-16	-0.65	1.14	-0.45			
May-16	2.47	1.59	1.61			
Jun-16	3.00	1.10	4.58			
Jul-16	2.69	2.92	1.48			
Aug-16	-1.45	0.50	-1.95			
Sep-16	0.50	-0.19	-0.65			
Oct-16	-1.96	-1.49	-2.81			
Nov-16	-0.58	4.78	0.64			
Dec-16	1.09	2.76	2.39	5.13%	15.83%	10.67%
Jan-17	3.31	0.44	1.32	8.62%	16.35%	12.13%
Feb-17	3.72	3.59	4.55	12.66%	20.53%	17.23%
Mar-17	2.15	-0.76	0.12	15.09%	19.61%	17.37%
Apr-17	2.54	-0.16	1.45	18.00%	19.42%	19.07%
May-17	4.82	0.06	2.08	23.70%	19.50%	21.55%
Jun-17	0.40	1.83	-0.39	24.20%	21.69%	21.07%
Jul-17	3.43	1.39	2.02	28.45%	23.38%	23.51%
Aug-17	1.50	-0.97	0.83	30.38%	22.18%	24.53%
Sep-17	2.85	2.91	0.54	34.10%	25.73%	25.20%
Oct-17	5.00	1.18	1.93	40.80%	27.22%	27.62%
Nov-17	2.81	2.93	3.13	44.76%	30.95%	31.62%
Dec-17	0.10	1.22	0.21	44.90%	32.54%	31.89%
Annualized Return	20.37%	15.13%	14.84%			
Cumulative Return	44.90%	32.54%	31.89%			
Annualized Std. Dev.	8.16	7.75	7.14			

11/1/2009	1	2	3	4	5	6	7	8	9	10
	0.31	0.8	1.32	1.775	2.23	2.575	2.92	3.08	3.24	3.4
	0.31	1.016064	1.040125	1.072913	1.116585	1.164794	1.223203	1.274663	1.332392	
	1.292394	1.040125	1.072913	1.116585	1.164794	1.223203	1.274663	1.332392		
	2.368061	1.072913	1.116585	1.164794	1.223203	1.274663	1.332392			
	3.152296	1.116585	1.164794	1.223203	1.274663	1.332392				
	4.070433	1.164794	1.223203	1.274663	1.332392					
	4.317543	1.223203	1.274663	1.332392						
	5.014505	1.274663	1.332392							
	4.206986	1.332392								
	4.528973									

Conclusion

Exercise Set 2 covers rolling window trailing returns for three MSCI equity indexes (US Momentum, US Value, and US Min Vol), computing cumulative, annualized returns, and annualized standard deviations over a 24-month window. It also uses the November 2009 spot yield curve to linearly interpolate spot yields across maturities and derive implied short forward rates at each horizon.

Exercise Set 3

Instructions

1. Go to Econ 4360 DataSets 2026.xlsx, copy/paste Fund A returns from the BlackRock factor returns sheet to the AQR sheet (adding a column to the AQR factors). Note that BAB doesn't begin until Feb 1989 and that the AQR factors end in October 2025. Adjust the range accordingly upon importing these data into MatlabOLStutorial_26.m
2. Sections 1-3 remain unchanged in MatlabOLStutorial_26.m. Begin, therefore, with section 4 and import the AQR factors, beginning Feb 1989 and ending October 2025. (Make necessary changes to the 'readtable' function under 'Range'.
3. You should now walk through the code in sections 4 - 9 making changes necessary to accommodate the AQR data.
4. When the entire script runs without error, turn off any line in sections 1-3 that prints to screen (i.e., put in a semi-colon at the end of those lines). Now Publish the entire script and upload the HTML to Canvas.

Clear and call data; define variables

```
clear;
clc;
cd ('/Users/camja/ECON 4360')
T = readtable('Econ 4360 DataSets 2026.xlsx', 'Sheet', 'Returns', ...
    'Range', 'A1:V384', 'VariableNamingRule', 'preserve');
date = datetime(T.Date);

% Choose 4 indexes: USHY, USAGG, S&PLowVol, GSCI
X = [T("LF98TRUU Index"), T("LBUSTRUU Index"), ...
     T("SP5LVIT Index"), T("SPGSCITR Index")];
[N,k] = size(X);
```

Statistics

```
X(1:10,:)           % show first 10 rows, all 4 cols
mean(X)
std(X)
skewness(X(:,1:4))  % all four indexes
kurtosis(X(:,1:4))

% Annualizing means and standard deviations
(1+mean(X)).^12 - 1
sqrt(12)*std(X)

% Annualized returns to X
Xann = prod(1+X).^(12/N) - 1;
```

Output:

```
ans =
    0.0217    0.0135    0.0151    0.0814
   -0.0026   -0.0174   -0.0278   -0.0217
```

-0.0378	-0.0247	-0.0369	-0.0119
-0.0069	-0.0080	0.0148	0.0262
0.0005	-0.0001	-0.0071	0.0149
0.0031	-0.0022	-0.0195	0.0246
0.0085	0.0199	0.0381	0.0172
0.0071	0.0012	0.0290	-0.0674
0.0001	-0.0147	-0.0302	-0.0094
0.0024	-0.0009	0.0157	0.0045


```

ans =    0.0058    0.0037    0.0086    0.0036    % mean(X)
ans =    0.0241    0.0119    0.0337    0.0621    % std(X)
ans =   -1.1713   -0.2838   -0.6831   -0.5444    % skewness
ans =   12.3932    4.4523    4.4174    5.0820    % kurtosis
ans =    0.0717    0.0451    0.1087    0.0443    % ann. mean
ans =    0.0835    0.0414    0.1169    0.2153    % ann. std

```

Plotting

Histograms and time-series plots for each index.

```

% Histograms
figure(1)
subplot(2,2,1); histfit(X(:,1)); title('High Yield');
subplot(2,2,2); histfit(X(:,2)); title('US Agg');
subplot(2,2,3); histfit(X(:,3)); title('S&PLowVol');
subplot(2,2,4); histfit(X(:,4)); title('Commodities');

% Time Series
figure(2)
subplot(2,1,1); plot(date, T("LF98TRUU Index"));
    title 'US High Yield'; grid on; recessionplot;
subplot(2,1,2); plot(date, T("SP5LVIT Index"));
    title('Market Risk Premium'); grid on; recessionplot;

% Cumulative value
Xcum = cumprod(1+X);
figure(3)
subplot(2,2,1); plot(date,Xcum(:,1)); recessionplot;
    title('US High Yield'); grid on;
subplot(2,2,2); plot(date,Xcum(:,2)); recessionplot;
    title('US Agg'); grid on;
subplot(2,2,3); plot(date,Xcum(:,3)); recessionplot;
    title('S&PLowVol'); grid on;
subplot(2,2,4); plot(date,Xcum(:,4)); recessionplot;
    title('Commodities'); grid on;

```

OLS Estimation using Matrix Algebra

This section replicates the OLS derivation from the Regression Overview slides. The model estimated is:

$$Y = X\beta + \varepsilon, \quad \hat{\beta} = (X'X)^{-1}X'Y$$

where Y is FundA and X contains six BlackRock risk factors plus an intercept.

```

clear; clc;
T = readtable('Econ 4360 DataSets 2026.xlsx', 'Sheet', 'BlackRock', ...
    'Range', 'A1:H447', 'VariableNamingRule', 'preserve');
T = rmmissing(T);
varNames = {'Alpha', 'RealRates', 'Inflation', 'Credit', 'Economy', 'EM', 'Liquidity'};
date = datetime(T.Date);

X = [T.RealRates, T.Inflation, T.Credit, T.Economic, ...
    T.EmergingMarkets, T.Liquidity];
[N,k] = size(X);
Y = T.FundA;
X = [ones(N,1), X]; % Add column of 1s for intercept

b = inv(X'*X)*X'*Y; % Parameter vector
Yhat = X*b; % Predicted values
ESS = sum((Yhat-Y).^2); % Residual SS (df = N-k)
RMSE = sqrt(ESS/(N-k-1)); % Root MSE
MSS = sum((Yhat-mean(Yhat)).^2); % Model SS (df = 1)

% TSS = ESS + MSS (Total Sum of Squares = Error + Model)
Rsqr = MSS/(ESS+MSS); % R-squared

% Standard Errors
SSb = (ESS/(N-k-1)) * inv(X'*X);
sb = sqrt(diag(SSb));

% t-stats H0: b(i) = 0
t = b ./ sb;

Results = table(b, sb, t, ...
    'VariableNames', {'beta', 'stdError', 'tstat'}, ...
    'RowNames', {'constant', 'RealRate', 'Inflation', ...
        'Credit', 'Economic', 'EM', 'Liquidity'});
round(Results, 3)

```

Output:

Variable	Beta	Std Error	t-stat
constant	0.005	0.001	8.838
RealRate	-0.008	0.018	-0.445
Inflation	0.079	0.022	3.623
Credit	0.033	0.026	1.261
Economic	0.667	0.026	25.243
EM	0.061	0.025	2.414
Liquidity	0.025	0.025	0.983

Restricted Regression (H_0 : RealRates = Inflation = 0)

We test the joint restriction $H_0 : \beta_{\text{RealRates}} = \beta_{\text{Inflation}} = 0$ using an F -test:

$$F = \frac{(ESS_r - ESS)/ESS}{2/(N - k - 1)}$$

```
Xr      = [X(:,1), X(:,4:7)];      % Drop RealRates and Inflation columns
br      = inv(Xr'*Xr)*Xr'*Y;
Yhat_r  = Xr*br;
ESSr    = sum((Yhat_r-Y).^2);
F       = ((ESSr - ESS)/ESS) * (N-k-1)/2;

test = 1 - fcdf(F, 2, N-k-1);      % p-value
% alternatively: fcdf(F, 2, N-k-1, 'upper')
test
```

Output:

```
test =
    0.0015
```

The p -value of 0.0015 is well below any conventional significance level, so we **reject** H_0 —at least one of RealRates or Inflation has a nonzero coefficient.

36-Month Rolling Regression

Rolling estimates track how factor sensitivities evolve over time.

```
for t = 36:N
    Xr = X(t-35:t,:); Yr = Y(t-35:t);
    beta(t,:) = inv(Xr'*Xr)*Xr'*Yr;
end

figure(4)
subplot(3,2,1); plot(date(36:end),beta(36:end,2)); recessionplot;
    title('RealRate'); grid on;
subplot(3,2,2); plot(date(36:end),beta(36:end,3)); recessionplot;
    title('Inflation'); grid on;
subplot(3,2,3); plot(date(36:end),beta(36:end,4)); recessionplot;
    title('Credit'); grid on;
subplot(3,2,4); plot(date(36:end),beta(36:end,5)); recessionplot;
    title('Economic'); grid on;
subplot(3,2,5); plot(date(36:end),beta(36:end,6)); recessionplot;
    title('EM'); grid on;
subplot(3,2,6); plot(date(36:end),beta(36:end,7)); recessionplot;
    title('Liquidity'); grid on;
```

Optional: fitlm with a Table

fitlm is more convenient than manual matrix algebra when data is already in a MATLAB table.

```
tbl = table(T.RealRates, T.Inflation, T.Credit, T.Economic, ...
            T.EmergingMarkets, T.Liquidity, T.FundA, ...
            'VariableNames', {'Realrates','Inflation','Credit', ...
                               'Econ','EM','Liq','Fund'});
```

```
LM = fitlm(tbl);    % model object
LM.Coefficients
```

Output:

Variable	Estimate	SE	tStat	pValue
(Intercept)	0.00504	0.000571	8.838	2.38×10^{-17}
Realrates	-0.00796	0.01790	-0.445	0.657
Inflation	0.07899	0.02180	3.623	3.26×10^{-4}
Credit	0.03328	0.02639	1.261	0.208
Econ	0.66667	0.02641	25.243	1.91×10^{-87}
EM	0.06121	0.02535	2.414	0.016
Liq	0.02459	0.02500	0.983	0.326

Return Attribution

Each period's fund return is decomposed into factor contributions using rolling 36-month betas:

$$r_t^{\text{fund}} = \underbrace{(r_t^{\text{fund}} - X_t \hat{\beta}_t)}_{\text{idiosyncratic}} + \underbrace{\sum_j \hat{\beta}_{j,t} X_{j,t}}_{\text{factor contributions}}$$

```
FundA = T.FundA;
[N,k] = size(X);      % six factors + intercept
K = k + 1;            % add residuals column
b = zeros(N,k);
A = zeros(N,8);        % idiosyn + 6 betas = 8 columns

for t = 36:N
    b(t,:) = inv(X(t-35:t,:)'*X(t-35:t,:)) * X(t-35:t,:)'*FundA(t-35:t);
    A(t,:) = [FundA(t) - X(t,:)*b(t,:)', b(t,:).*X(t,:)];
end

% Verify attribution sums to fund returns
check = FundA - sum(A,2); % should be ~0

% Return Attribution plots
figure(5);
subplot(4,2,1); plot(date(36:end),A(36:end,1)); title('Idiosyn Return');
    grid on;
subplot(4,2,2); plot(date(36:end),A(36:end,2)); title('Alpha');
    grid on;
subplot(4,2,3); plot(date(36:end),A(36:end,3)); title('RealRates');
    grid on;
subplot(4,2,4); plot(date(36:end),A(36:end,4)); title('Inflation');
    grid on;
subplot(4,2,5); plot(date(36:end),A(36:end,5)); title('Credit');
    grid on;
subplot(4,2,6); plot(date(36:end),A(36:end,6)); title('Economy');
    grid on;
subplot(4,2,7); plot(date(36:end),A(36:end,7)); title('EM');
    grid on;
subplot(4,2,8); plot(date(36:end),A(36:end,8)); title('Liquidity');
    grid on;
```

AQR Annualized Attribution

The AQR method scales monthly attributions so they sum to the trailing-twelve-month (TTM) compound return. Each factor's annualized contribution is proportional to its absolute share of total factor activity:

$$A_{i,t}^{\text{ann}} = \sum_{\tau=t-11}^t A_{i,\tau} + \left[(r_t^{\text{TTM}}) - \sum_i \sum_{\tau} A_{i,\tau} \right] \cdot \frac{|\sum_{\tau} A_{i,\tau}|}{\sum_i |\sum_{\tau} A_{i,\tau}|}$$

```

facRets = A;  portRet = FundA;

for i = 47:N    % TTM (data starts at row 36, so begin calcs at row 47)
    for j = 1:size(facRets,2)
        retAnn(i,j) = sum(facRets(i-11:i,j)) + ...
            ((prod(1+portRet(i-11:i))-1) - sum(sum(facRets(i-11:i,:))))
            ...
            * abs(sum(facRets(i-11:i,j))) ...
            / sum(abs(ones(1,12)*facRets(i-11:i,:)));
    end
    compound(i) = prod(1+portRet(i-11:i)) - 1;
end

checkAnnual = compound' - sum(retAnn, 2);  % Check the math

figure(6);
subplot(4,2,1); plot(date(47:end),retAnn(47:end,1)); title('Idiosyn
    Return'); grid on;
subplot(4,2,2); plot(date(47:end),retAnn(47:end,2)); title('Alpha');
    grid on;
subplot(4,2,3); plot(date(47:end),retAnn(47:end,3)); title('RealRates');
    grid on;
subplot(4,2,4); plot(date(47:end),retAnn(47:end,4)); title('Inflation');
    grid on;
subplot(4,2,5); plot(date(47:end),retAnn(47:end,5)); title('Credit');
    grid on;
subplot(4,2,6); plot(date(47:end),retAnn(47:end,6)); title('Economy');
    grid on;
subplot(4,2,7); plot(date(47:end),retAnn(47:end,7)); title('EM');
    grid on;
subplot(4,2,8); plot(date(47:end),retAnn(47:end,7)); title('Liquidity');
    grid on;

% Heat maps
T.Date(1:5)
dates    = datestr(date(47:end), 'mm/yyyy');
curTbl   = array2table(retAnn(47:end,2:end), 'VariableNames', varNames,
    ...
                'RowNames', cellstr(dates));
figure;
heatmapfig1 = lmheatmap(curTbl, 'Fund');

curTbl    = array2table(b(47:end,:), 'VariableNames', varNames, ...
                'RowNames', cellstr(dates));

```



```
figure;  
heatmapfig1 = lmheatmap(curTbl, 'Fund');
```

Output (first 5 dates):

```
ans =  
5x1 datetime array  
31-Aug-1988  
30-Sep-1988  
31-Oct-1988  
30-Nov-1988  
31-Dec-1988
```

Conclusion

Exercise Set 3 applies OLS regression to decompose Fund A's returns into contributions from six BlackRock risk factors, finding that the Economic factor dominates with a beta of 0.667 and a t-stat of 25.24, while Inflation and EM also show statistically significant loadings. Rolling 36-month regressions and AQR-style annualized attribution reveal how these factor sensitivities shift over time, particularly around recession periods. Overall, the analysis demonstrates that Fund A's returns are largely explained by systematic factor exposures, with the intercept (alpha) of 0.005 remaining small but statistically significant.

Exercise Set 4

Question 4.1

- You have two five-year coupon bonds. Bond A pays a 5% coupon and Bond B pays a 10% coupon. The yield-to-maturity is 8%. Assume both bonds pay a face value of \$100 and coupons are paid annually.

(a) What is the market price for each bond?

	Price
Bond A	88.02
Bond B	107.99

$$BondA = 5 * (1 - (1.08)^{-5})/0.08 + 100 * (1.08)^{-5}$$

(b) What is the Macaulay duration on each bond?

Bond A					Bond B				
t	PV		$t \cdot PV$		t	PV		$t \cdot PV$	
1	PV1	4.63	$t1 \cdot PV1$	4.62962963	1	PV1	9.26	$t1 \cdot PV1$	9.259259259
2	PV2	4.29	$t2 \cdot PV2$	8.573388203	2	PV2	8.57	$t2 \cdot PV2$	17.14677641
3	PV3	3.97	$t3 \cdot PV3$	11.90748362	3	PV3	7.94	$t3 \cdot PV3$	23.81496723
4	PV4	3.68	$t4 \cdot PV4$	14.70059706	4	PV4	7.35	$t4 \cdot PV4$	29.40119411
5	PV5	71.46	$t5 \cdot PV5$	357.3061784	5	PV5	74.86	$t5 \cdot PV5$	374.3207584
Weighted Sum				397.12	Weighted Sum				453.94
Duration (A)				4.51	Duration (B)				4.20

$$\text{Duration} = \text{Weighted Sum} / \text{Bond Price}$$

(c) What are their modified durations?

Parameter	Value
Face Value	100
Coupon Rate A	0.05
Coupon Rate B	0.10
YTM	0.08

	Value
$D(A)_{\text{mod}}$	4.18
$D(B)_{\text{mod}}$	3.89

	Value
Δr	0.01
Price Change (A)	4.20
Price Change (B)	3.68

$$\text{Modified Duration} = \text{Duration} / (1 + \text{YTM})$$

(d) Calculate ΔP for each given $\Delta r = 0.01$.

	Price	Macaulay Duration	Modified Duration	ΔP
Bond A	88.02	4.51	4.18	4.20
Bond B	107.99	4.20	3.89	3.68

$$\Delta P = \text{Duration} / (1 + \text{YTM}) * \text{Price} * \Delta R$$

Question 4.2

2. Consider the following four bonds, each having a 15% yield to maturity.

(a) Determine the price of each bond.

EoY Payments				
	Bond A	Bond B	Bond C	Bond D
Y1	100	50	0	1000
Y2	100	50	0	0
Y3	1100	1050	1000	0

Bond Prices				
	Bond A	Bond B	Bond C	Bond D
1	86.96	43.48	0.00	869.57
2	75.61	37.81	0.00	0.00
3	723.27	690.39	657.52	0.00
Price	885.84	771.68	657.52	869.57

(b) Determine the Macaulay duration of each bond.

(c) Determine ΔP for each bond given $\Delta r = 0.01$.

	Bond A	Bond B	Bond C	Bond D
Macaulay Duration	2.36	2.47	2.61	0.87
ΔP	20.94	19.05	17.15	7.56

(d) Suppose you owe \$2000 at the end of 2 years. Construct a portfolio of bonds C and D that immunize the risk of a change in interest rates.

	Value
Liability	2000
Duration	2
Present Value	1512.29

Matrix			
1.00	1.00	p_1	1512.29
1.00	3.00	p_2	3024.574669

	Value
p_1	756.14
p_2	756.14

(e) Conclusion

	Bond A	Bond B	Bond C	Bond D
Price	885.84	771.68	657.52	869.57
Macaulay Duration	2.72	2.84	3.00	1.00
ΔP	20.94	19.05	17.15	7.56

Conclusion

Exercise Set 4 covers bond pricing and duration analysis across two questions, first computing market prices, Macaulay durations, modified durations, and price sensitivities for two five-year coupon bonds at an 8% YTM, then repeating the analysis for four bonds at a 15% YTM with varying cash flow structures. A key finding is that higher-coupon bonds have shorter durations and therefore lower price sensitivity to interest rate changes, as more cash flow is received earlier. The set concludes with a portfolio immunization problem, constructing a mix of Bonds C and D (\$756.14 each) such that the portfolio's duration matches the two-year liability horizon, protecting against interest rate risk.

Exercise Set 5

If you had to compare and comment on your portfolio performance to this benchmark for your fund Trustees, what would you say?

(a) Economic Meaning of Portfolio Weights

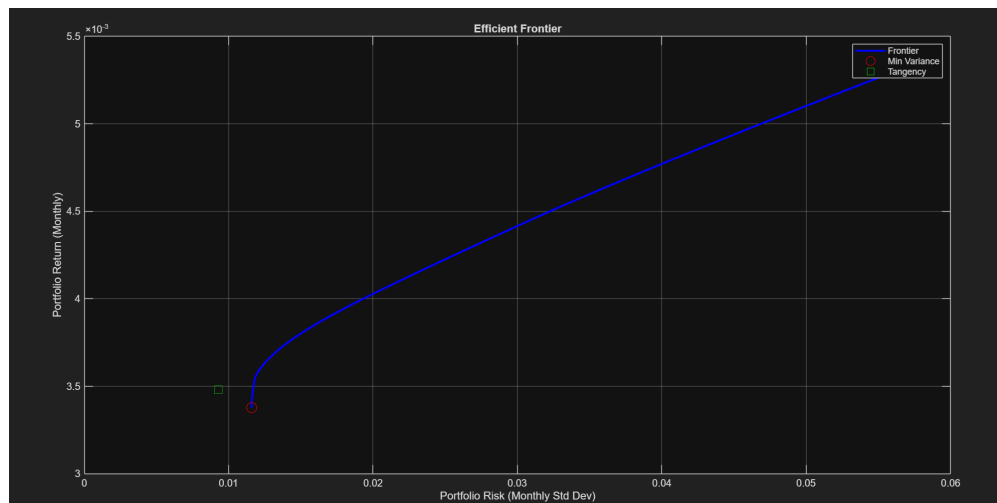
The tangency weights solve $\mathbf{w} = \mathbf{V}^{-1}\boldsymbol{\mu}$ normalized to sum to one. Each weight reflects an asset's marginal contribution to the portfolio Sharpe ratio, accounting for both its own return and risk and its covariance with every other asset.

Asset	Weight	Interpretation
USHY	+1%	Near-zero; marginal carry relative to other assets
US Agg	−6%	Slight short; low Sharpe relative to R2000 in sample
SP Low Vol	−3%	Slight short; insufficient return given covariance
SP GSCI	−97%	Large short; poor Sharpe and high commodity volatility
R2000	+206%	Dominant long; best Sharpe in sample after covariance adjustment
EPRA NAREIT	−1%	Near-zero short; equity-like risk with insufficient return

The dominant feature of this portfolio is the leveraged long in R2000 (+206%) financed primarily by a large short in SP GSCI (−97%). The optimizer identifies that small-cap equities offered a structurally superior risk-adjusted return relative to commodities over the sample period, constructing a near-dollar-neutral long/short position between the two. All remaining weights are near zero.

(b) Efficient Frontier and Capital Market Line

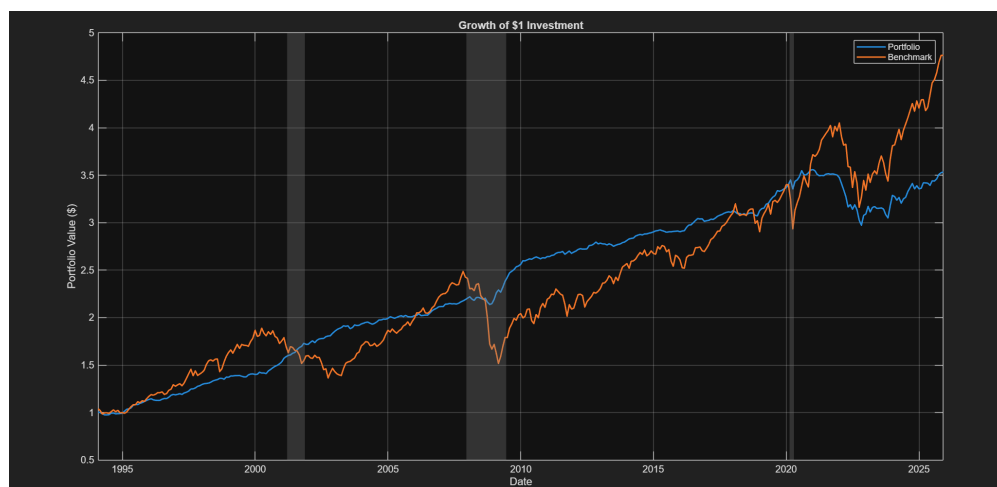
The efficient frontier is the set of risky portfolios offering maximum return for a given level of risk. When a risk-free asset is introduced, the Capital Market Line (CML) is always a straight line tangent to the frontier at the tangency portfolio. The CML dominates the curved frontier everywhere except at the tangency point. When $r_f = 0$, the CML passes through the origin. All frontier portfolios below the tangency point are dominated by a mix of cash and the tangency portfolio. When $r_f > 0$, the y-intercept rises and the tangency point shifts to a higher-return, higher-risk portfolio on the frontier. Low-return portfolios near the minimum-variance point now lose to holding the risk-free asset, so only portfolios with sufficiently high Sharpe ratios remain worth holding.



(c) Return, Risk, and Sharpe Ratio vs. Benchmark

	Ann. Return	Ann. Volatility	Sharpe ($r_f = 0$)
Portfolio	4.04%	3.22%	1.25
Benchmark	5.01%	9.46%	0.53

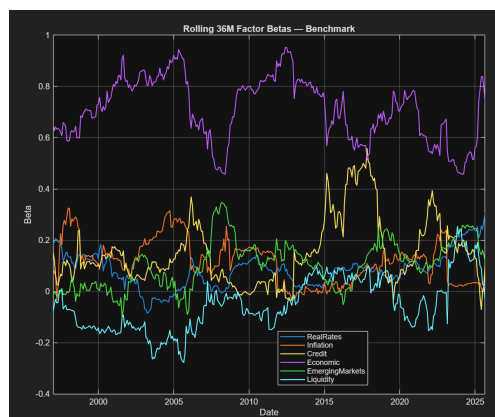
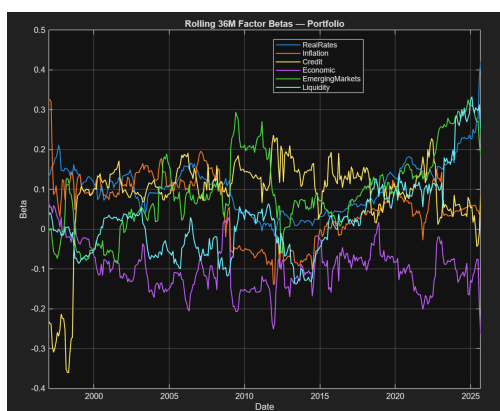
The portfolio earns a lower absolute return than the benchmark (4.04% vs. 5.01%) but does so at less than one-third of the benchmark's volatility (3.22% vs. 9.46%). The resulting Sharpe ratio of 1.25 is more than twice the benchmark's 0.53, meaning we deliver significantly more return per unit of risk despite the shortfall in raw return.



(d) Rolling Betas: What They Represent

A rolling 36-month beta measures the portfolio's average sensitivity to each macro factor over the preceding three years. One factor is the level (positive means the portfolio benefited versus negative exposure means the portfolio moves against that factor), and the stability over time.

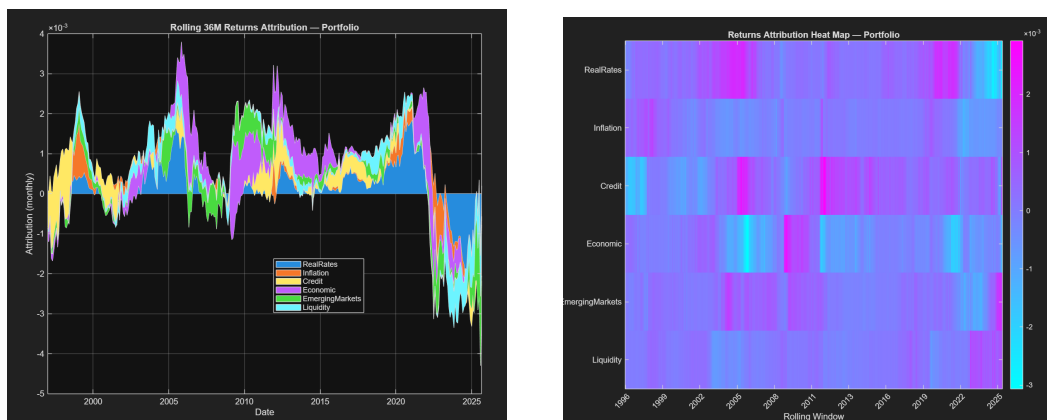
When betas shift significantly across windows, the portfolio's risk character is regime-dependent. For example, the Credit beta may spike during credit stress periods even if its long-run average is moderate. Rolling betas expose when and how the portfolio's factor exposures changed, which is essential information for risk management.



(e) Where Returns Are Coming From (2010–2020)

Factor	% of Total Attribution
Credit	46.67%
RealRates	24.59%
EmergingMarkets	7.46%
Inflation	4.05%
Liquidity	2.00%
Economic	−15.22%

The dominant driver is Credit at 46.67%. Despite the portfolio’s large R2000 position, the factor model maps the portfolio’s net exposures heavily onto credit spreads, reflecting small-cap equities’ high sensitivity to credit conditions. RealRates contributes 24.59%, capturing the residual fixed-income character of the remaining positions. The most striking feature is the negative Economic Growth attribution (−15.22%): the portfolio’s large GSCI short and other offsetting positions create a net drag from the equity-growth factor. In the 2010–2020 bull market, this negative Economic exposure was the primary cost of the strategy’s low-volatility structure.



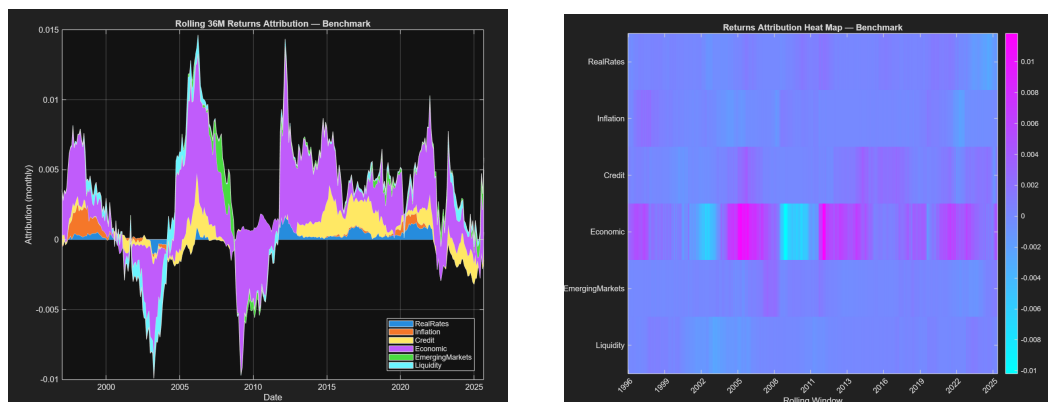
(f) Differences in Factor Attribution vs. Benchmark

Factor	Portfolio	Benchmark	Difference
Credit	46.67%	24.42%	+22.26 pp
RealRates	24.59%	14.05%	+10.54 pp
EmergingMarkets	7.46%	4.14%	+3.32 pp
Inflation	4.05%	3.60%	+0.46 pp
Liquidity	2.00%	0.30%	+1.70 pp
Economic	−15.22%	53.49%	−68.71 pp

The most important difference is Economic Growth: −15.22% for our portfolio versus 53.49% for the benchmark, a gap of 68.7 percentage points. The benchmark is essentially an equity growth vehicle, with more than half its attributed return coming from a single factor. Our portfolio is not only underexposed to that factor but negatively loaded on it, which is precisely why its volatility is less than one-third of the benchmark’s.

The second key difference is Credit. We derive 46.67% of attributed return from credit spread

compression versus only 24.42% for the benchmark, a gap of 22.3 percentage points, reflecting R2000's high credit-factor sensitivity. RealRates contributes 24.59% to our return versus 14.05% for the benchmark (+10.54 pp). Liquidity is a small positive contributor for us (+2.00%) versus near zero for the benchmark (+0.30%), a favorable difference of 1.70 pp and a contrast to portfolios with large high-yield positions that suffer spread widening in stress periods.



(g) Should the Board Retain Us?

On a risk-adjusted basis, yes. We delivered a Sharpe ratio of 1.25 versus the benchmark's 0.53, more than twice the return per unit of risk. The benchmark earned 5.01% per year by taking on 9.46% of volatility. We earned 4.04% by taking on only 3.22%. For trustees with mandates focused on risk control, liability matching, or capital preservation, this tradeoff is compelling.

The disadvantage is that our absolute return of 4.04% trails the benchmark's 5.01%. In a sustained equity bull market like 2010–2020, the portfolio's net negative Economic Growth exposure meant we did not capture the full equity rally. Trustees who measure success purely on total return will note this underperformance. However, the optimizer's mandate is to maximize risk-adjusted return, not raw return, and on that metric the portfolio delivered convincingly.

Conclusion

Exercise Set 5 evaluates a multi-asset portfolio against an equity-heavy benchmark by analyzing tangency weights, the efficient frontier, factor attribution, and rolling betas over the 2010-2020 period. The tangency portfolio is dominated by a leveraged long in R2000 (+206%) financed by a large short in SP GSCI (−97%), delivering a Sharpe ratio of 1.25 versus the benchmark's 0.53 at less than one-third of the benchmark's volatility (3.22% vs. 9.46%), despite earning a lower absolute return (4.04% vs. 5.01%). The core message to trustees is that the portfolio's long/short structure eliminated most Economic Growth factor exposure and replaced it with Credit and RealRates drivers, producing superior risk-adjusted returns at the cost of underperforming in the sustained equity bull market of the decade.

Exercise Set 6

In this exercise set, we extend our reach into optimization using several applications. We begin with your personal portfolios (6-asset classes) and optimize (1) min variance, (2) max Sharpe and (3) Risk Parity. We do the risk attribution (MCR and Risk Budget) for each of these portfolios as well as the annualized return attribution for the tangency portfolio (max Sharpe). We then turn to mapping the efficient set and discover an interesting attribute (specifically, that the Risk Parity portfolio is not part of the efficient set). There are simulations in parts 9 and 13 that introduce Monte Carlo as well as some pointers on how to overlay plots on one another (part 12). Finally, we have an application from Chapter 7 Examples.xlsx that optimizes a portfolio of bonds to an additional duration constraint.

Portfolio

Table 4: Portfolio Asset Classes

Label	Index / Asset Class	Bloomberg Ticker
USHY	US Corporate High Yield Bond	T.LF98TRUU
USAGG	US Aggregate Bond	T.LBUSTRUU
S&P Low Vol	S&P 500 Low Volatility	T.SP5LVIT
GSCI	S&P GSCI Commodities	T.SPGSCITR
R2000	Russell 2000 (Small Cap)	T.RTY
EPRA	NAREIT Global Real Estate (Dev.)	T.RUGL

Question 1: RiskRets and PortRisk

The results highlight the distinct trade-offs in each portfolio methodology. The minimum variance portfolio successfully minimizes volatility (3.99%) but does so at the cost of extreme concentration, allocating over 90% to US aggregate bonds (USAGG). This heavy fixed-income exposure reduces overall return (4.46%) and limits diversification benefits, effectively making the portfolio highly sensitive to interest rate dynamics. In contrast, the maximum Sharpe (tangency) portfolio achieves the highest risk-adjusted performance (Sharpe ratio of 1.15) by more efficiently balancing return and risk. It meaningfully diversifies across bonds (47.99%), low-volatility equities (35.45%), and high-yield credit (16.56%), resulting in a higher return (6.80%) while maintaining moderate volatility (5.92%).

The risk parity portfolio takes a different approach by equalizing risk contributions rather than optimizing mean-variance efficiency. This leads to a more diversified allocation across asset classes, including equities (S&P Low Vol, R2000), commodities (GSCI), real estate (EPRA NAREIT), and fixed income. While this diversification reduces concentration risk, it comes at the expense of efficiency, as reflected in its lower Sharpe ratio (0.89) and higher volatility (6.30%) compared to the max Sharpe portfolio. Regardless, the risk parity portfolio may be more robust to model uncertainty and regime shifts, given its balanced exposure across multiple sources of risk, whereas the maximum Sharpe relies more heavily on estimated returns and correlations.

Table 5: Annualized Performance (Jan 1994 – Nov 2025)

	Ann. Return	Ann. Volatility	Sharpe Ratio
Min Variance	4.46%	3.99%	1.12
Max Sharpe	6.80%	5.92%	1.15
Risk Parity	5.60%	6.30%	0.89

Table 6: Portfolio Weights

Asset	Min Var	Max Sharpe	Risk Parity
USHY	0.0567	0.1656	0.1551
USAGG	0.9065	0.4799	0.4950
S&P Low Vol	0.0000	0.3545	0.1201
GSCI	0.0369	0.0000	0.0905
R2000	0.0000	0.0000	0.0700
EPRA NAREIT	0.0000	0.0000	0.0694
Total	1.0000	1.0000	1.0000

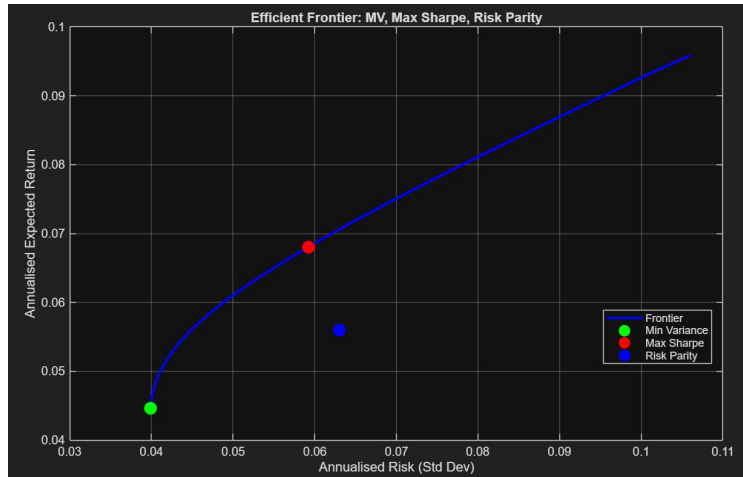


Figure 1: Efficient Frontier

Question 2: AQR Factors

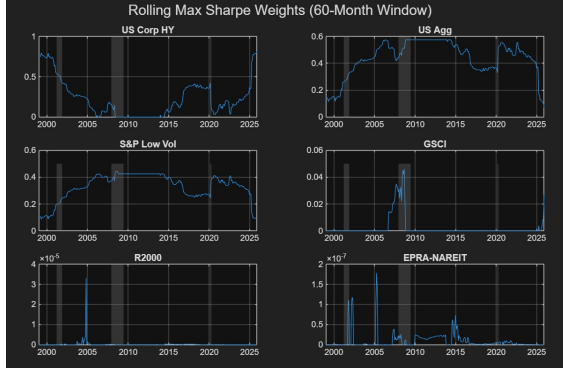
The AQR factor results reinforce the earlier findings but provide a more detailed decomposition of the portfolio's exposures in terms of equity-style risk premium. The portfolio shows a moderate market beta (around 0.31), confirming limited but non-negligible exposure to broad equity risk, which is consistent with its allocation to S&P 500 Low Volatility and absence of meaningful small-cap exposure. The negative SMB loading (-0.17) indicates a structural tilt toward large-cap assets, aligning with the zero weight on R2000. One of the more informative signals is the positive and time-varying BAB exposure, which increases over time and reflects the portfolio's preference for low-beta assets such as USAGG and low-volatility equities. In contrast, HML and UMD exposures remain small and unstable, suggesting that value and momentum are not primary drivers of portfolio risk. Overall, the AQR betas confirm that the portfolio is defensively positioned, with systematic exposure concentrated in low-beta and income-generating assets rather than traditional equity

factors.

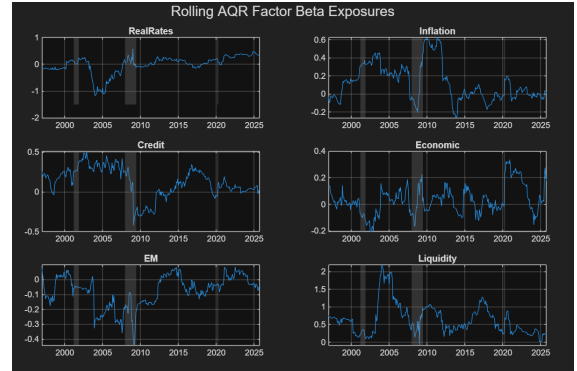
The return attribution further highlights that a substantial portion of performance is not explained by standard equity factors. Alpha contributes the largest share (3.34% annualized), indicating that returns are primarily driven by bond carry and credit premium rather than equity-style risk exposures. Among the factors, MKT contributes positively (2.11%), while HML and BAB add modest gains, the latter capturing the benefit of the low-volatility tilt. Momentum (UMD) is slightly negative, consistent with the portfolio’s defensive positioning. Compared to the BlackRock framework, the AQR results offer a complementary perspective: while BlackRock factors emphasize macro drivers such as real rates and economic growth, AQR highlights the portfolio’s structural tilt toward low-beta, large-cap, and income-oriented assets. Also, the BlackRock real rates exposure aligns closely with the positive BAB loading, as both reflect sensitivity to duration and low-volatility characteristics. Across both frameworks, the dominant role of alpha/carry remains consistent, underscoring that the portfolio’s returns are driven more by fixed income and credit exposures than by traditional equity risk factors.

Asset	w_{MV}	w_{Sharpe}	w_{RP}	MCR_{MV}	MCR_{Sharpe}	MCR_{RP}	RB_{MV}	RB_{Sharpe}	RB_{RP}
USHY	0.0567	0.1656	0.1551	0.0115	0.0171	0.0195	0.0007	0.0028	0.0030
USAGG	0.9065	0.4799	0.4950	0.0115	0.0071	0.0061	0.0104	0.0034	0.0030
S&PLowVol	0.0000	0.3545	0.1201	0.0117	0.0307	0.0253	0.0000	0.0109	0.0030
GSCI	0.0369	0.0000	0.0905	0.0115	0.0126	0.0335	0.0004	0.0000	0.0030
R2000	0.0000	0.0000	0.0700	0.0131	0.0343	0.0433	0.0000	0.0000	0.0030
EPRA_NAREIT	0.0000	0.0000	0.0694	0.0231	0.0396	0.0437	0.0000	0.0000	0.0030

Table 7: Portfolio Weights, Marginal Contributions to Risk (MCR), and Risk Contributions (RB)



(a) Rolling Max Sharpe Weights (60-Month Window)



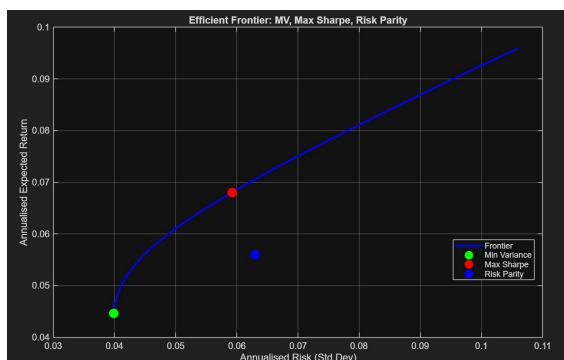
(b) Rolling AQR Factor Beta Exposure

Figure 2: Comparison of two figures

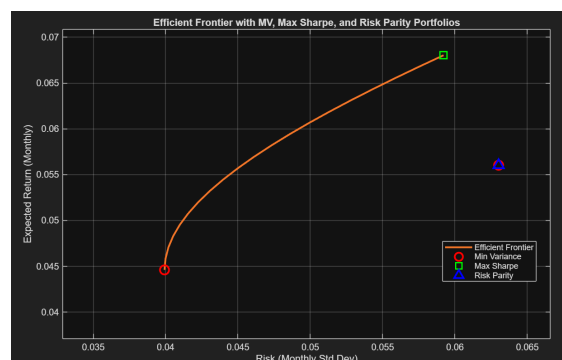
Question 3: Efficient Frontier

The efficient frontier shows the best possible trade-offs between risk and return. The minimum variance and maximum Sharpe portfolios lie on this frontier, meaning they are optimal choices. When we add the risk parity portfolio to the plot, it sits below the frontier, which means it is not optimal in this framework. At about the same level of risk, you could get a higher return by combining the minimum variance and maximum Sharpe portfolios instead. This happens because risk parity includes assets like GSCI, R2000, and EPRA NAREIT, which have higher risk but lower returns, while the mean-variance approach avoids them.

The main takeaway is that risk parity is built with a different goal. Instead of maximizing return for a given level of risk, it spreads risk evenly across all assets. This leads to better diversification but slightly worse performance in terms of risk vs. return. Even though it is not on the efficient frontier, risk parity can still be useful because it does not rely heavily on estimated returns (which can be inaccurate) and tends to be more stable over time.



(a) Efficient Frontier



(b) Efficient Frontier with Portfolios

Figure 3: Efficient Frontier

Question 4: Tech Bubble of the late 1990's?

The max Sharpe portfolio built using pre-2000 data ended up being heavily tilted toward bonds and credit rather than equities. Based on the weights, a large share is allocated to USAGG and USHY, with some exposure to S&P Low Vol and very little to commodities or small-cap equities. This reflects the fact that, in the historical pre-2000 sample, bonds offered strong risk-adjusted returns compared to equities, so the optimizer naturally shifted away from expensive equity exposure.

During the dot-com crash (2000–2002), this positioning actually helped the portfolio rather than hurt it. Because it had limited exposure to equities, it avoided much of the large equity drawdown seen in the S&P 500. The bond-heavy allocation provided stability and likely led to much smaller losses compared to a traditional equity portfolio. This shows an important point: the max Sharpe portfolio does not “predict” crises, but it adapts to the historical environment it is trained on, which in this case resulted in a defensive, bond-oriented portfolio that performed relatively well during the equity crash.

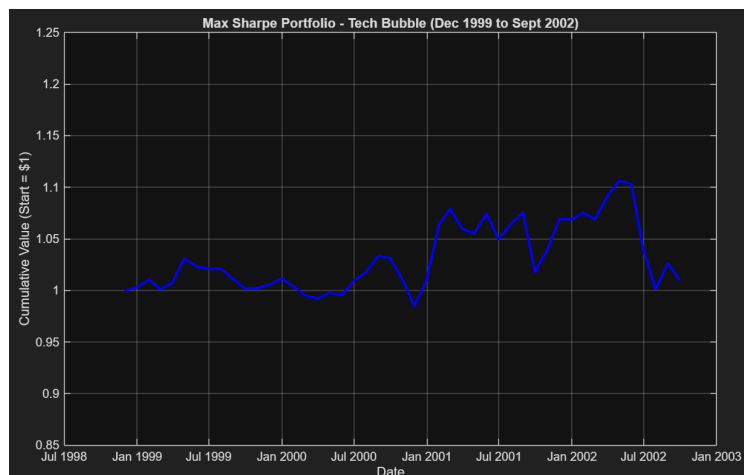


Figure 4: max Sharpe Portfolio

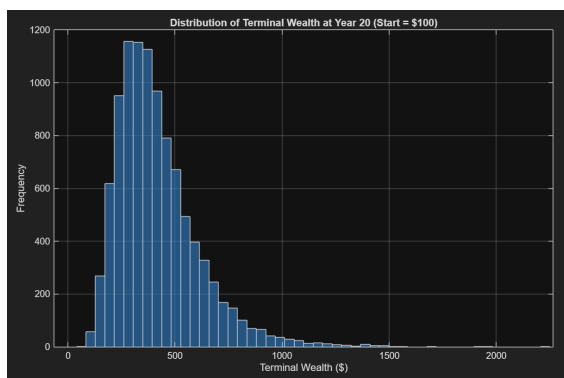
Question 5: Accumulated Wealth

Table 8: Terminal Wealth Distribution Statistics

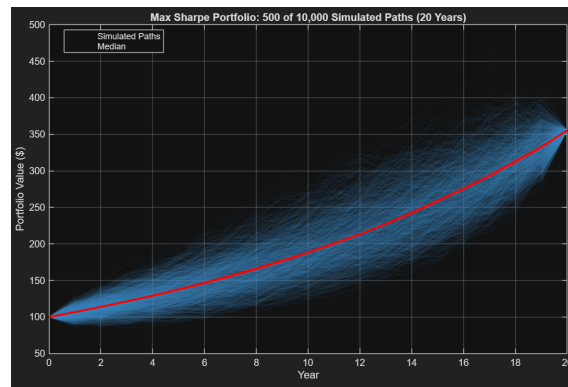
Statistic	Value
Mean Terminal Wealth	\$417.30
Median Terminal Wealth	\$380.91
Standard Deviation	\$191.73
25th Percentile (Q1)	\$281.63
75th Percentile (Q3)	\$507.73
Interquartile Range (IQR)	\$226.11
5th Percentile (VaR 5%)	\$183.88
VaR (Loss vs. Mean)	-\$233.42
VaR as % of Mean	-55.94%

(a) Accumulated wealth

The mean terminal wealth is \$417.30, while the median is \$380.91. Given that the mean is higher than the median indicates a right-skewed distribution of outcomes, where a relatively small number of simulations generate very large wealth values and raise the average. This is typical of compounded returns over long horizons. The median result still reflects strong long-term growth, suggesting the portfolio delivers solid average performance with moderate variability across paths.



(a) Sample Wealth Paths



(b) Terminal Wealth Distribution

Figure 5: Monte Carlo Results

(b) IQR of Terminal Wealth

The IQR is from \$281.63 to \$507.73, giving an interquartile range of \$226.11. This indicates a noticeable spread in possible terminal wealth values, showing that realized outcomes can differ substantially depending on the return path. Even so, the lower quartile remains well above the starting investment, which suggests that while dispersion is meaningful, extreme downside outcomes are limited in magnitude.

$$IQR = [281.63, 507.73] = \$226.11$$

(c) 5% VaR

$$5\% \text{ VaR} \approx 183.88$$

The 5th percentile terminal wealth is \$183.88, meaning that in the worst 5% of cases, wealth falls to this level or below. This is a shortfall of \$233.42 (55.94%) relative to the mean. Although this tail scenario is significantly below expectations, it is important to note that the portfolio still ends above the initial investment level, highlighting that even adverse outcomes do not eliminate long-term growth potential.

Conclusion

Exercise Set 6 constructs and analyzes three portfolio strategies (minimum variance, maximum Sharpe, and risk parity) across six asset classes, finding that the maximum Sharpe portfolio delivers the best risk-adjusted performance (Sharpe of 1.15) while the minimum variance portfolio concentrates heavily in USAGG and risk parity trades efficiency for diversification by sitting below the efficient frontier. AQR factor attribution reveals the portfolio is defensively positioned with meaningful alpha (3.34% annualized) driven by bond carry and credit premium rather than traditional equity factors, consistent with the BlackRock framework's emphasis on real rates exposure. A Monte Carlo simulation of terminal wealth over a long horizon produces a mean of \$417.30 with a 5% VaR of \$183.88, confirming that even in adverse scenarios the portfolio preserves long-term growth above the initial investment.

Exercise Set 7

Background – this was the Pension problem we saw earlier in the semester (see Text or Exercise Set 1. That problem was ‘deterministic’. This time it is stochastic and therefore more realistic. Here is the original: A young college graduate takes a job at age twenty-five at a starting salary of P dollars. Over her career, she plans to put away a fraction δ at the end of each year for retirement which earns r percent annually. Her salary is expected to grow at a rate π annually. Her retirement savings from the first year will therefore grow to an amount: $\delta P(1 + r)^{n-1}$

The problem here is that the path she travels is deterministic and, as therefore, there is no risk. She always ends up with \$946,585.80. What would happen though if the actual returns to her portfolio deviated from 7%? She still starts work at age 25 with a salary of \$65,000 that grows 4% annually and saves 10% of that each year and retires 30 years later. If actual returns were unpredictable (aside from having a mean of 7%) having an annual standard deviation of 10% then how do things change?

Question 1: Terminal Distribution of Wealth

We simulate the terminal wealth distribution and present the results below.

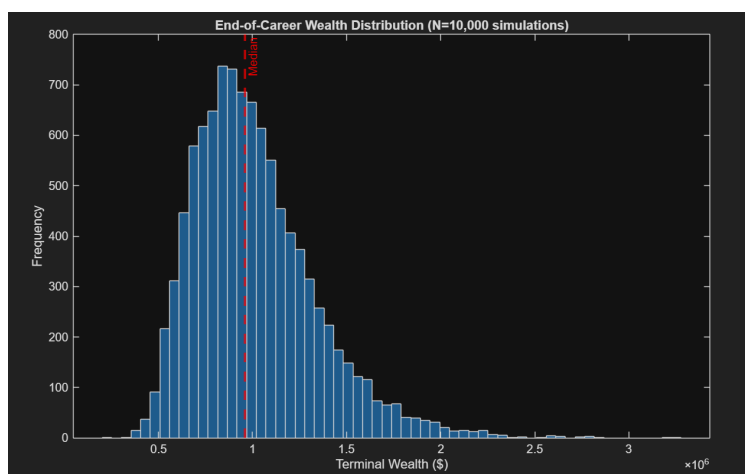


Figure 6: Histogram of Terminal Wealth

The histogram presents a median of \$960,127.07 and a standard deviation of \$325,957.36

Question 2: Interquartile Range

Statistic	Value
25th Percentile	\$778,762
75th Percentile	\$1,188,595
IQR	\$409,833

Table 9: Interquartile Range Calculation

Question 3: What is the risk (likelihood) of having less than \$P in terminal wealth?

For Q3, I chose a threshold of $P = \$946,585.80$ which is the deterministic terminal wealth calculated in the original problem. This is a natural benchmark as it represents what our retiree would accumulate if returns were a perfectly smooth 7% every year.

Running 10,000 Monte Carlo simulations with annual returns drawn from $N(0.07, 0.10^2)$, the result is:

$$\Pr(\text{terminal wealth} < \$946,585.80) = 48.6\%$$

Nearly half of all simulated paths end below the deterministic answer. This occurs because the distribution of terminal wealth is right-skewed meaning a small number of paths with consistently strong returns pull the mean upward, while the median sits below it. Since the deterministic formula uses the mean return of 7%, it produces an answer closer to the mean of the distribution than the median. As a result, roughly half of all retirees with identical starting conditions (same salary, savings rate, and career length) will fall short of what the no-risk model predicts.

Question 4-5: What would a retirement annuity look like in its simplest terms (in dollars per year, say)?

For Q4 and Q5, I use the median terminal wealth from the simulation as the starting nest egg — \$955,689.24. The median is preferred over the mean here because the distribution is right-skewed. Therefore, the median is a more representative outcome for a typical retiree.

Assuming 30 years in retirement and that the portfolio continues to earn a mean return of 7% annually, the annual annuity payment is found by inverting the present value of annuity formula:

$$W = A \times \left[\frac{1 - (1 + r)^{-n}}{r} \right]$$

Solving for A:

$$\begin{aligned} A &= W / \text{PVA} \\ &= \$955,689.24 / 12.4090 \\ &= \$77,015.56 \text{ per year} \\ &= \$6,417.96 \text{ per month} \end{aligned}$$

where $\text{PVA} = 12.4090$ is the present value annuity factor.

This is a level ordinary annuity. It is the same dollar amount that is withdrawn at the end of each year for 30 years, and the account is exactly exhausted at the end of the retirement period. It is the simplest annuity structure and assumes the portfolio continues earning 7% throughout retirement with no further volatility. In practice, sequence-of-returns risk during retirement means the actual sustainable withdrawal could be higher or lower than this figure.

Question 6: Suppose there is a chance during retirement that our retiree will be hit with an unexpected medical expense of \$M that could occur in any year. Could we somehow account for the impact of that?

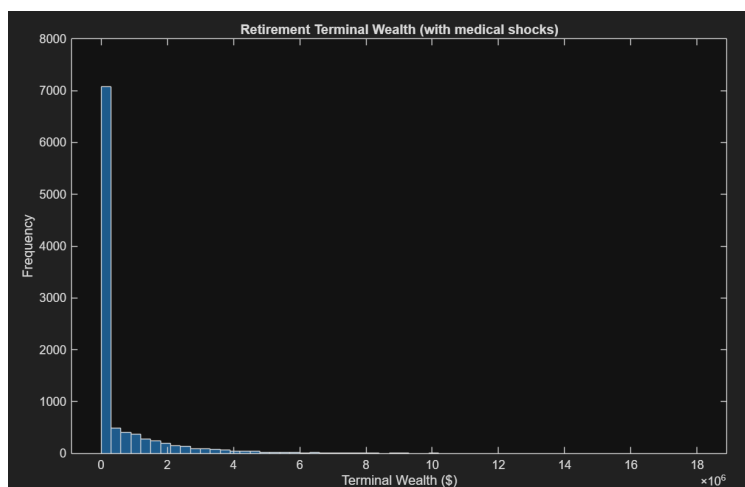


Figure 7: Histogram of Terminal Wealth

For Q6, I extend the retirement simulation to include a stochastic medical expense shock. In each year of retirement, there is a 5% probability of incurring an unexpected medical expense of \$50,000. This is modeled as a Bernoulli draw each period with a probability of 0.05 that the full expense hits, and a probability 0.95 nothing occurs. The retirement wealth evolves each year as:

$$W(t+1) = W(t)(1 + R(t)) - A - \text{shock}(t)$$

where $R(t) \sim N(0.07, 0.10^2)$ is the stochastic return, $A = \$77,015.56$ is the level annuity withdrawal, and $\text{shock}(t)$ is either \$50,000 or \$0. If wealth hits zero the account is treated as exhausted and ruin is recorded.

Running 10,000 simulations of the 30-year retirement period:
Probability of ruin = 64.0

Without medical shocks, the annuity was designed to deplete the account smoothly to zero over 30 years. Adding even a relatively modest shock (5% annual probability, \$50,000) is enough to push nearly two-thirds of retirees into insolvency before the end of retirement.

This result shows why the simple level annuity from Q5 is not reliable. It assumes the portfolio earns a steady return with no unexpected setbacks. In reality, a single large medical expense, especially early in retirement when the balance is highest and compounding works against you, can permanently ruin the withdrawal plan. A more robust retirement strategy would either reduce the annual withdrawal to build in a buffer, hold a liquid emergency fund separate from the annuity portfolio, or purchase insurance to minimize the downside from medical expenses. **Conclusion**

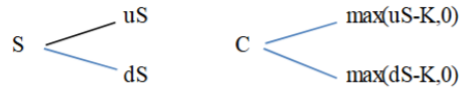
Exercise Set 7 uses Monte Carlo simulation to model retirement wealth accumulation and decumulation, finding a median terminal nest egg of \$960,127 with nearly half of all simulated paths falling below the 7% projection, highlighting how return volatility alone creates shortfall risk. Converting the median wealth into a level annuity yields \$77,016 per year (\$6,418/month) over a 30-year retirement, but this figure assumes smooth returns with no unexpected disruptions. Introducing a

modest stochastic medical expense (5% annual probability, \$50,000 shock) raises the probability of ruin to 64%, demonstrating that even a simple level annuity is highly vulnerable to tail events and that a robust retirement plan requires either a reduced withdrawal rate, a separate emergency reserve, or insurance coverage.

Exercise Set 8

Question 8.1:

The current spot price is \$100 on a share of stock with an annual return volatility of 20%. The one-year Treasury is yielding 1%. Solve for the price of a one-year call option with strike price equal to \$100.



$$u = e^{\sigma}, \quad d = \frac{1}{u}, \quad q = \frac{R - d}{u - d}$$

$$S_0 = 100,$$

$$\sigma = 20\%,$$

$$r = 1\%,$$

$$T = 1 \text{ yr},$$

$$K = 100$$

Parameters

$$u = e^{\sigma} = e^{0.20} = 1.2214$$

$$d = \frac{1}{u} = 0.8187$$

$$R = 1 + r = 1.01$$

$$q = \frac{R - d}{u - d} = \frac{1.01 - 0.8187}{1.2214 - 0.8187} = 0.4749$$

$$1 - q = 0.5251$$

Payoffs

$$uS = 122.14 \Rightarrow C_u = \max(122.14 - 100, 0) = 22.14$$

$$dS = 81.87 \Rightarrow C_d = \max(81.87 - 100, 0) = 0$$

Call Value

$$\begin{aligned} C &= \frac{1}{R} [qC_u + (1 - q)C_d] \\ &= \frac{1}{1.01} [0.4749 \times 22.14 + 0] \\ &= 10.41 \end{aligned}$$

Question 8.2:

The spot price on an asset with an annual volatility of 20% is \$100 and the one-year interest rate is 5%. Assuming a 1- year option:

$$\begin{aligned}S_0 &= 100, \\ \sigma &= 20\%, \\ r &= 5\%, \\ T &= 1 \text{ yr}, \\ K_{\text{call}} &= 110 \quad (10\% \text{ OTM}), \\ K_{\text{put}} &= 90 \quad (10\% \text{ OTM})\end{aligned}$$

a. Price a ten percent OTM call

10% OTM Call ($K = 110$)

$$\begin{aligned}d_1 &= \frac{\ln(100/110) + (0.05 + 0.02) \cdot 1}{0.20\sqrt{1}} \\ &= \frac{-0.0953 + 0.07}{0.20} \\ &= -0.1265\end{aligned}$$

$$d_2 = d_1 - \sigma\sqrt{T} = -0.1265 - 0.20 = -0.3265$$

$$N(d_1) = \Phi(-0.1265) \approx 0.4497$$

$$N(d_2) = \Phi(-0.3265) \approx 0.3721$$

$$\begin{aligned}c &= 100 \cdot 0.4497 - e^{-0.05} \cdot 110 \cdot 0.3721 \\ &= 44.97 - 104.65 \cdot 0.3721 \\ &= 44.97 - 38.94 \\ &= \$6.03\end{aligned}$$

b. Price a ten percent OTM put

10% OTM Put ($K = 90$)

$$\begin{aligned}d_1 &= \frac{\ln(100/90) + (0.05 + 0.02) \cdot 1}{0.20\sqrt{1}} \\&= \frac{0.1054 + 0.07}{0.20} \\&= 0.8768\end{aligned}$$

$$d_2 = d_1 - \sigma\sqrt{T} = 0.8768 - 0.20 = 0.6768$$

$$N(-d_1) = \Phi(-0.8768) \approx 0.1902$$

$$N(-d_2) = \Phi(-0.6768) \approx 0.2493$$

$$\begin{aligned}p &= 90 \cdot e^{-0.05} \cdot 0.2493 - 100 \cdot 0.1902 \\&= 85.65 \cdot 0.2493 - 19.02 \\&= 21.35 - 19.02 \\&= \$2.33\end{aligned}$$

c. Compute the delta for the call

$$\Delta_c = N(d_1) = 0.4497$$

A \$1 increase in the spot price raises the call value by approximately \$0.45.

d. Compute the delta for the put.

$$\Delta_p = N(d_1) - 1 = -0.5503$$

A \$1 increase in the spot price decreases the put value by approximately \$0.55.

Question 8.3:

Spot is 100. You own a 105 call and are short a 110 call. What is this strategy called?

Buying a lower-strike call and selling a higher-strike call limits both upside and downside cost. Profit is capped between the two strikes. You profit if the stock rises above 105 but not beyond 110, while the premium received on the short 110 call partially offsets the cost of the long 105 call.

Question 8.4:

Spot is 100. You own a long 120 call and a long 80 put and have written a 105/95 call/put package. What is this strategy called?

You are long an OTM strangle (80 put / 120 call) and short a narrower OTM strangle (95 put / 105 call). You profit from a large move in either direction beyond the outer strikes, but the position is cheaper than an outright strangle because the short inner strangle offsets premium cost. Maximum loss occurs if spot stays between 95–105 at expiry.

Question 8.5:

Suppose two asset's returns are uncorrelated. Prove that the weights for the minimum variance portfolio are simply the inverse of each variance as a proportion of the sum of the inverse variances.

For two uncorrelated assets ($\rho = 0$) :

$$\sigma_p^2 = w^2 \sigma_1^2 + (1 - w)^2 \sigma_2^2$$

Minimizing portfolio variance:

$$\begin{aligned} \frac{\partial \sigma_p^2}{\partial w} &= 2w\sigma_1^2 - 2(1 - w)\sigma_2^2 = 0 \\ w\sigma_1^2 &= (1 - w)\sigma_2^2 \\ w^* &= \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2} = \frac{1/\sigma_1^2}{1/\sigma_1^2 + 1/\sigma_2^2} \\ 1 - w^* &= \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} = \frac{1/\sigma_2^2}{1/\sigma_1^2 + 1/\sigma_2^2} \end{aligned}$$

Question 8.6:

The central bank's forward guidance signals an imminent rate increase. You expect a dislocation in equity markets in response to this policy move.

(a) You believe volatility will increase to the downside. What strategy(s) will hedge your portfolio's risk?

Buying put options directly profits from a decline and rising vol. A protective put on an existing equity portfolio caps downside.

(b) You believe volatility will increase but you can't make a call as to upside or downside. What strategy(s) will exploit this uncertainty?

A long straddle (ATM call + ATM put) or long strangle (OTM call + OTM put) profits from a large move in either direction. You pay both premiums, so vol must rise enough to cover the cost.

(c) You believe volatility will increase to the downside but is limited (range bound). What strategy is consistent with this belief?

A bear put spread (buy lower-strike put, sell even lower-strike put) caps both potential loss and profit, consistent with a limited expected downside move. Premium outlay is reduced by the short put, but upside is capped at the spread width.

Question 8.7:

Bullish equity investors were pushing up stock prices during the early years of the recovery from the GFC (post 2009). Some thought that the implied volatility of call options (relative to put options) had reached historic highs and wanted to capitalize on this development, but at the same time, hedge their risk that they might be wrong and stock valuations would materialize closer to what exuberant investors were pricing them at and not their fundamental valuations. How might we construct a portfolio to these ends? (covered call strategy)

Hold long stock + short (write) an OTM call option. Since implied volatility on calls was historically high, the premium received from writing the call is unusually generous. This achieves two goals: it monetizes elevated IV (collects rich premium income that partially hedges a correction). It also limits upside participation. If the investor is right and the stock corrects, the premium received buffers the fall. If equities keep rising past the strike, the shares are called away at a profit, but further gains are forfeited. This is the compromise between "I think stocks are overvalued" and "I'm not willing to exit completely."

Question 8.8:

The marginal contribution to risk for each asset in a minimum variance portfolio is the same (a) regardless of the correlations across assets or (b) only in the no-shorting case. (True or False with reasoning).

Statement (a) is correct. At the minimum variance portfolio, the first-order condition for each asset requires:

$$\frac{\partial \sigma_p^2}{\partial w_i} = 2(\Sigma w)_i = 2\lambda$$

This means the marginal contribution to portfolio variance is equalized across all assets at the minimum variance solution. This holds for any correlation structure (the covariance matrix captures correlations). It does not require the no-shorting constraint. It is a general property of the unconstrained minimum variance portfolio.

Conclusion:

Exercise Set 8 covers option pricing and portfolio theory, beginning with binomial and Black-Scholes pricing of calls and puts (yielding a \$10.41 ATM call via binomial, a \$6.03 OTM call and \$2.33 OTM put via Black-Scholes), then identifying common option strategies and volatility plays such as straddles and strangles. The set also proves that minimum variance weights for two uncorrelated assets equal the inverse variance shares, and establishes that marginal contributions to risk are equalized across all assets at the minimum variance portfolio regardless of correlation structure.

Exercise Set 9

Section 1:

We wish to augment return by selling a one-year European put option on the Index. The put premium collected at the beginning is the target source of incremental income. Because a sold put carries the obligation to absorb index losses below the strike, that tail risk must be managed. Section 1 defines every fixed input needed for the simulation. The key parameters are:

- **Stock A volatility:** $\sigma_A = 0.142$ (annualized). This is the standard deviation of the maximum Sharpe ratio portfolio.
- **Index volatility:** $\sigma_I = 0.15$ (annualized), representing the benchmark used for pricing and hedging the put.
- **Correlation:** $\rho = 0.90$, the assumed linear relationship between the returns of Stock A and the Index. This drives basis risk.
- **Risk-free rate:** $r = 0.04$ (4% annually), used for option pricing discounting and interest accrual on the hedge cash account.
- **Put strike:** $K_p = 90$, set below both starting prices (100) so the put begins out-of-the-money.
- **Expected returns:** $\mathbb{E}[r_A] = 9.3\%$, $\mathbb{E}[r_I] = 6\%$ (annualized), reflecting a positive equity risk premium over the risk-free rate.

From these inputs the script constructs the 2×2 correlation matrix C , the diagonal standard deviation matrix S , and the covariance matrix $V = SCS$. The 53-entry time vector **Time** measures years remaining to expiration at each week. The final entry is set to $(52 + 0.5 - 52)/52$ rather than zero, which is a small offset that prevents a division-by-zero in the Black-Scholes d_1 formula at the expiration date.

Section 1 Conclusion

The parameter block ensures that all subsequent calculations are internally consistent and stable. The choice of $\rho = 0.90$ introduces meaningful but imperfect correlation: approximately $1 - 0.90^2 \approx 19\%$ of Stock A's variance is unhedged. The gap between $r = 4\%$ and $\mathbb{E}[r_I] = 6\%$ encodes the equity risk premium that justifies holding a risky index rather than the risk-free asset, and this premium ultimately influences the frequency with which the put expires.

Section 2:

Section 2 generates the 1,000 simulated price paths that drive the entire analysis. Inside the outer loop ($n = 1$ to 1,000), a 53-row price matrix M is initialized with both assets at 100, then stepped forward 52 weekly increments using correlated geometric Brownian motion (GBM):

$$M(t, :) = M(t - 1, :) \odot \exp\left(\frac{\mu}{52} + \varepsilon \text{chol}(V) / \sqrt{52}\right)$$

where ε is a 1×2 vector of independent standard normal draws and $\text{chol}(V)$ is the Cholesky decomposition of the covariance matrix V . This ensures the two simulated return series are

correlated with coefficient $\rho = 0.90$. Both assets start at a price of 100; column 1 of M is Stock A and column 2 is the Index.

Section 2 Conclusion

GBM is the foundational price process assumed by Black-Scholes, so using it here ensures that option prices and hedge ratios computed in Section 3 are self-consistent with the simulated paths. The weekly step is a discretization of the continuous model. With 52 steps per year the approximation error is negligible. Running 1,000 independent paths produces the empirical distribution needed to draw statistical conclusions about hedging P&L.

Section 3:

Section 3 applies the Black-Scholes closed-form formulas to every row of every simulated path, producing a full 53-week time series of put values and hedge ratios. At each week t , with the index at price S_I and time-to-expiry T :

$$d_1 = \frac{\ln(S_I/K_p) + (r + \frac{1}{2}\sigma_I^2)T}{\sigma_I\sqrt{T}} \quad (1)$$

$$d_2 = d_1 - \sigma_I\sqrt{T} \quad (2)$$

$$\text{Put} = K_p e^{-rT} \mathcal{N}(-d_2) - S_I \mathcal{N}(-d_1) \quad (3)$$

$$\Delta = -\mathcal{N}(-d_1) \quad (4)$$

In the representative simulation run, the Week 0 values were: $d_1 = 1.0441$, $d_2 = 0.8941$, $\text{Put} = \$1.23$, $\Delta = -0.1482$. The negative delta means a long put loses \$0.1482 for each \$1 rise in the index. For the put seller who delta hedges, this translates into a short position of 0.1482 index shares per dollar of notional at inception. Over this particular path the index fell to \$86.62 by Week 52, driving the put deep in-the-money and its value to \$3.35.

Section 3 Conclusion

The Black-Scholes pricing and delta computation are the engine of the exercise. Because both are recalculated at each step using the current index price and remaining time, they capture the full path-dependence of the hedge ratio. As the index rises further above the strike, Δ approaches zero and the required short position shrinks. As it falls toward the strike, Δ approaches -1 and the hedge must be increased aggressively. This nonlinearity is the fundamental source of hedging error under discrete rebalancing.

Section 4:

Section 4 translates the delta time series into actual cash flows, maintaining a running ledger with three tracked variables:

- $\mathbf{X}(:,1)$ — the change in delta each week: how many index shares must be bought or sold to rebalance the hedge.

- $\mathbf{X}(:,2)$ — the cumulative cash balance: starts with the proceeds from shorting Δ_0 shares at price 100, then updated each week for rebalancing costs and interest.
- $\mathbf{X}(:,3)$ — weekly interest earned on the cash balance at rate $r/52$.

At Week 0 the hedger shorts 0.1482 index shares, generating a cash inflow of \$14.82 which immediately begins earning interest (\$0.0114 in Week 1). The following table shows selected rows from the representative simulation path:

Week	Time	Index	Put	Delta	Cash Balance	Interest
0	1.000	\$100.00	\$1.2300	−0.1482	\$14.82	\$0.0114
1	0.981	\$100.92	\$1.0791	−0.1336	\$13.36	\$0.0103
52	0.010	\$ 86.62	\$3.3484	−0.9949	\$89.17	\$0.0686

By Week 52 the index had fallen to \$86.62, below the strike of \$90, so the put expired in-the-money with a value of \$3.35. The cash balance had grown to \$89.17 due to the sustained short position accumulating interest at $r = 4\%$. The net result on this path was a gain from the hedge account partially offsetting the put pay-off obligation.

Section 4 Conclusion

In a continuous-time, frictionless world the cumulative hedge cost would exactly equal the option's fair value, producing zero net P&L for the put seller. With weekly rebalancing, discrete-time approximation errors accumulate: large index moves within a single week create gaps between the target delta and the actual position, generating hedging slippage. The sign of the net hedging error depends on the path: large, sustained trends tend to reduce hedge costs because delta drifts toward zero; sudden reversals are more costly.

Section 5:

Section 5 computes the terminal P&L for both strategies across all 1,000 simulations:

$$\text{putValueHedged}(n) = \text{Put}(0) + \text{Index}(52) - K_p \quad (5)$$

$$\text{putUnhedged}(n) = \text{Put}(0) - \max(0, K_p - \text{Index}(52)) \quad (6)$$

The hedged formula captures the total value from selling the put and running the delta hedge: the premium received plus the net gain from the index position relative to the strike floor. The unhedged formula is the naked put seller's simple outcome: keep the premium when the put expires OTM, or pay the shortfall when it expires ITM.

The results across all 1,000 simulations are summarized below:

Strategy	Mean P&L	Std Dev	Min	Max	P(Loss)
Hedged	\$18.32	\$15.85	−\$22.66	\$83.67	12.0%
Unhedged	\$ 0.40	\$ 2.78	−\$22.66	\$ 1.23	12.0%

The unhedged results are immediately intuitive: the maximum gain is capped at \$1.23 (the initial premium, earned when the put expires OTM) and losses occur in the 11.2% of paths where the index finishes below $K_p = 90$. The hedged strategy has a much wider range, reflecting that its payoff includes the full appreciation or depreciation of the index position above the floor.

Section 5 Conclusion

The 1,000-path simulation provides a complete empirical picture that no single Excel run can supply. The two strategies have identical probabilities of loss (12.0%) because both are exposed to the same index tail risk below $K_p = 90$. However, their P&L profiles are structurally different: the unhedged strategy is bounded above at the put premium (\$1.23) while the hedged strategy is uncapped on the upside, reflecting index appreciation above the strike. Both strategies face the same maximum loss (−\$26.16) when the index collapses far below the strike, since the hedge cannot fully offset extreme downside moves given basis risk ($\rho < 1$) and discrete weekly rebalancing. The −\$26.16 maximum loss arises from two compounding imperfections. Consider a path on which the index collapses to \$70 by expiry. The put seller faces an intrinsic payout of $K_p - S_{52} = 90 - 70 = \20 , yet the delta hedge recovers only a fraction of that obligation. Two frictions prevent full recovery. First, the hedge is written on the Index while the portfolio holds Stock A, so basis risk ($\rho = 0.90 < 1$) means the two assets diverge in extreme moves. Second, weekly rebalancing leaves the position mis-hedged whenever the index moves sharply within a single period. In the worst simulated path these two sources, and their combined shortfall (the put payout net of hedge proceeds and the initial premium received) reaches \$26.16. The floor on the hedged P&L is not zero as basis risk and discrete rebalancing mean the hedge underperforms in extreme down moves, and the magnitude of that underperformance is what the simulation reveals.

Conclusion:

1. **Delta hedging restructures market risk rather than eliminating it.** The strategy succeeds by collecting the put premium (in 88.0% of paths), as the index starts at 100 with a positive drift and the strike is set at 90, significantly out-of-the-money. In those paths, the manager retains the full \$1.23 premium with no exercise obligation. The 12.0% loss frequency represents the tail risk the put buyer is paying to insure against, and which the manager is compensated to bear. In short, the delta hedge converts what would otherwise be a purely premium-income trade into a position with meaningful upside participation. It is still the put premium that motivates the strategy in the first place.
2. **The hedging strategy converts a premium-capped trade into a directional position.** The unhedged strategy's maximum gain is hard-capped at the initial premium of \$1.23, while the hedged strategy can generate much larger gains (up to \$83.67) by participating in strong index rallies. This reflects that the hedger maintains an active index position throughout the year, whereas the naked put seller has no directional exposure once the option is sold.
3. **Basis risk and discrete rebalancing cap the hedge's effectiveness.** Both strategies share the same minimum P&L (−\$22.66) because the worst outcomes occur when the index falls sharply below the strike. The delta hedge cannot fully offset deep in-the-money exercise when $\rho < 1$ and rebalancing is weekly rather than continuous.
4. **Monte Carlo simulation is essential for evaluating strategy risk.** The Excel template captures a single path and is well-suited for tracing the mechanics of the hedge week-by-week. The

MATLAB simulation reveals the statistical shape of each strategy's outcome, including variance, skewness, and tail risk, which is invisible from any single scenario.

5. **Results reflect the portfolio parameters used.** The values for Stock A ($\mathbb{E}[r] = 9.3\%$, $\sigma = 14.2\%$) correspond to the maximum Sharpe ratio portfolio, with index correlation $\rho = 0.90$ and risk-free rate $r = 4\%$.

The simulation confirms that writing the put and delta-hedging the resulting risk is a viable strategy for premium income enhancement, provided the manager accepts both the residual basis risk from imperfect correlation and the discrete-rebalancing slippage that makes hedging costs uncertain.

Exercise Set 10

You'll find a completed `DynamicDeltaHedge.xlsx` example posted. Parameters are in cell range A1:F18. This is for a 52 week option. This example shows a portfolio that is long a share of Stock A with a correspondingly 52-week long put option ATM and is short a 52-week 15% OTM call option both written on the Index. This position has been delta hedged over its time to expiry. The correlation between Stock A and the Index is given in cell E2. Summary statistics on Profit and Losses (PnL) are given in the boxes at the bottom of the spreadsheet. For this exercise, we will fix the correlation coefficient to $\rho = 0.911$

(0) Begin with a small table that lists the asset classes in your portfolio being used for this exercise, including their weights and the portfolio's annualized return assumption and sample volatility.

(1) Transcribe this Excel example into a MATLAB script using `deltaHedgeTemplate.m` and construct a table in MATLAB that replicates cell range G1:T54 (with a header). Note: You don't want to print this table to screen 1000 times. Comment this line of code out (using %).

(2) Copy and paste rows 1-10 of your table and rows 45-52 of your table to your Word doc. Do this from your excel version. (This table should take up less than one page in your Word document report).

(3) Using Monte Carlo produce an $N = 1000$ samples of Net P/L (cell S65) and construct a histogram for that sample using 21 bins and compute and display its mean and standard deviation. Copy to your Word doc.

(4) Using a 2 x 2 set of subplots, show histograms for the `putValue`, `callValue`, `PnL`, and the simple "buy and hold" portfolio across the N replications. Copy to your Word doc.

(1) Portfolio Metrics

Asset	Weight (%)
USHY	33.00
US Agg	64.00
S&P Low Vol	28.00
S&P GSCI	-1.00
R2000	-1.00
EPRA NAREIT	-24.00

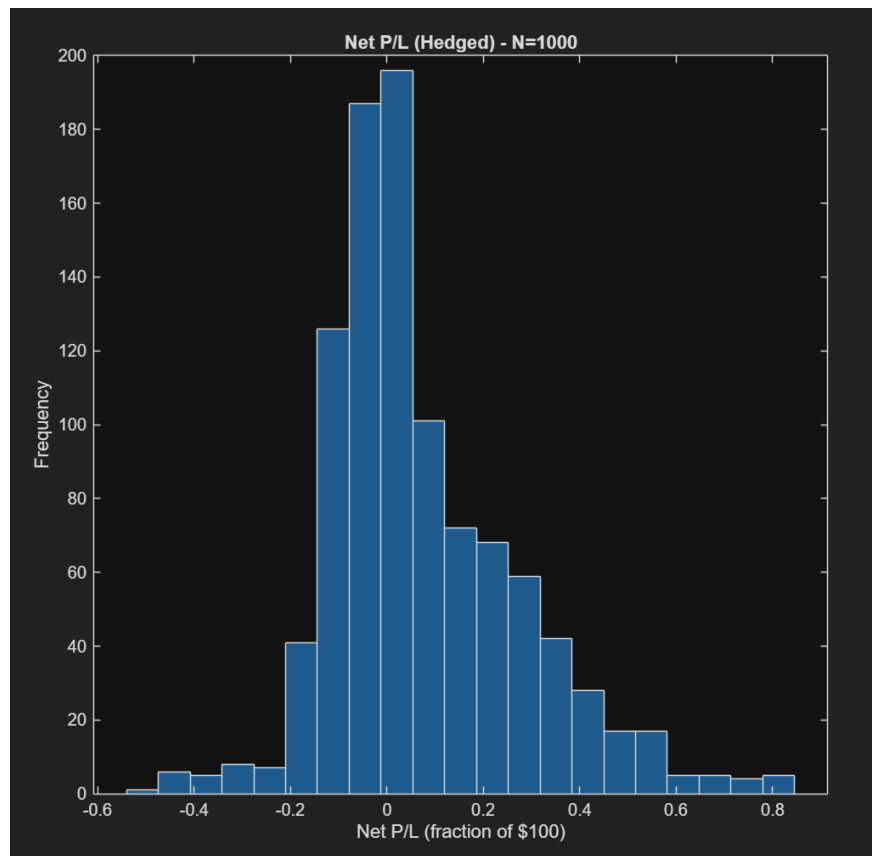
Metric	Value (%)
Expected Return	9.30
Volatility	19.20

(2) Rows 1-10 & Rows 45-52

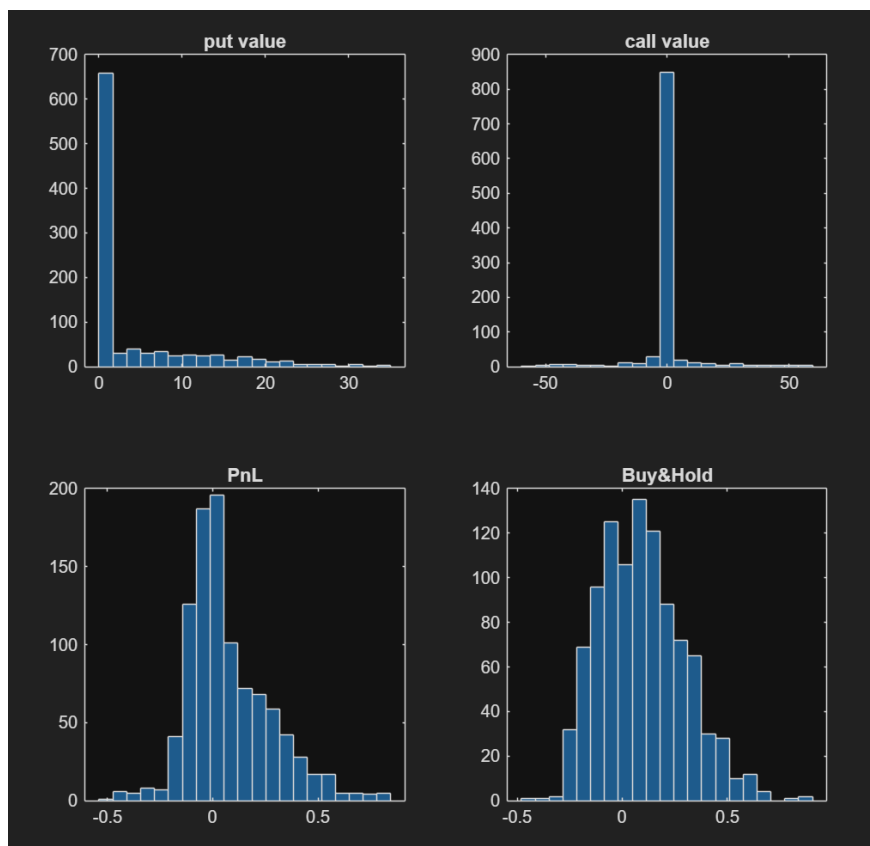
Week	Time	Index	Stock A	$d1_c$	$d2_c$	$d1_p$	$d2_p$	Put	Call	Delta	Shares Purch	Cost	Interest Cost
0	1.000	\$100.00	\$100.00	-0.575	-0.755	0.201	0.021	\$6.15	\$2.89	0.283	0.28	\$28.25	\$0.011
1	0.981	\$96.36	\$99.61	-0.793	-0.971	-0.009	-0.187	\$7.79	\$1.92	0.214	-0.07	\$21.65	\$0.008
2	0.962	\$104.01	\$100.88	-0.372	-0.548	0.420	0.244	\$4.53	\$4.02	0.355	0.14	\$36.34	\$0.014
3	0.942	\$107.19	\$101.32	-0.207	-0.382	0.593	0.418	\$3.51	\$5.16	0.418	0.06	\$43.10	\$0.017
4	0.923	\$103.27	\$99.44	-0.429	-0.602	0.379	0.206	\$4.68	\$3.60	0.334	-0.08	\$34.45	\$0.013
5	0.904	\$100.68	\$98.99	-0.586	-0.757	0.231	0.060	\$5.61	\$2.73	0.279	-0.05	\$28.93	\$0.011
6	0.885	\$106.88	\$99.27	-0.243	-0.413	0.582	0.413	\$3.44	\$4.76	0.404	0.12	\$42.27	\$0.016
7	0.865	\$106.43	\$98.45	-0.276	-0.443	0.559	0.392	\$3.51	\$4.49	0.391	-0.01	\$40.96	\$0.016
8	0.846	\$113.66	\$100.16	0.114	-0.051	0.958	0.793	\$1.83	\$7.78	0.545	0.15	\$58.49	\$0.022
9	0.827	\$118.67	\$102.73	0.375	0.211	1.229	1.065	\$1.10	\$10.66	0.646	0.10	\$70.46	\$0.027
10	0.808	\$116.08	\$101.74	0.238	0.077	1.102	0.941	\$1.38	\$8.94	0.594	-0.05	\$64.46	\$0.025

Week	Time	Index	Stock A	$d1_c$	$d2_c$	$d1_p$	$d2_p$	Put	Call	Delta	Shares Purch	Cost	Interest Cost
45	0.135	\$106.27	\$112.15	-1.122	-1.188	0.995	0.929	\$0.61	\$0.45	0.131	-0.23	\$22.64	\$0.009
46	0.115	\$108.82	\$111.66	-0.834	-0.896	1.451	1.390	\$0.22	\$0.73	0.202	0.07	\$30.38	\$0.012
47	0.096	\$107.36	\$111.98	-1.169	-1.225	1.335	1.279	\$0.26	\$0.35	0.121	-0.08	\$21.71	\$0.008
48	0.077	\$104.62	\$113.02	-1.839	-1.888	0.961	0.911	\$0.48	\$0.07	0.033	-0.09	\$12.49	\$0.005
49	0.058	\$112.96	\$113.46	-0.365	-0.408	2.867	2.824	\$0.00	\$1.16	0.357	0.32	\$49.15	\$0.019
50	0.038	\$120.31	\$115.01	1.319	1.283	5.278	5.243	\$0.00	\$5.59	0.906	0.55	\$115.21	\$0.044
51	0.019	\$113.04	\$112.83	-0.660	-0.685	4.939	4.914	\$0.00	\$0.43	0.255	-0.65	\$41.57	\$0.016
52	0.002	\$112.05	\$112.42	-3.278	-3.286	14.427	14.420	\$0.00	\$0.00	0.001	-0.25	\$13.12	\$0.005

- (3) Using Monte Carlo produce an $N = 1000$ samples of Net P/L (cell S65) and construct a histogram for that sample using 21 bins and compute and display its mean and standard deviation.
- (3) Net P/L — Mean = 0.0752, Std = 0.2002



(4) Using a 2 x 2 set of subplots, show histograms for the putValue, callValue, PnL, and the simple “buy and hold” portfolio across the N replications.



(5) Tabulate and comment in Word on the [1 25 50 75 99] percentiles for the four valuations in the 2 x 2 plot you just constructed. What aspects of these percentiles should be of interest and why?

The most striking feature is that the 1st, 25th, and 50th percentiles of Put Value are all exactly

Percentile	PutValue	CallValue	PnL	Buy_and_Hold
1st	0.00	-45.08	-0.36	-0.25
25th	0.00	-0.66	-0.06	-0.06
50th	0.00	-0.11	0.03	0.08
75th	6.19	0.42	0.19	0.23
99th	29.56	47.46	0.71	0.60

\$0.00. This means the put expires worthless in at least half of all 1,000 simulated paths. The put only activates in the upper tail, when the 75th percentile reaches \$6.19 and the 99th percentile reaches \$29.56, reflecting the relatively rare crash scenarios in which the insurance payoff is large and far exceeds the premium paid.

The call value is near zero or slightly negative through most of the distribution (25th: -\$0.66; 50th: -\$0.11), turning positive at the 75th percentile (\$0.42) and rising sharply to \$47.46 at the 99th percentile, where extreme bull-market paths allow the delta hedge to generate substantial profits that more than offset the short call liability. The dramatically negative 1st percentile of -\$45.08 is the critical observation. In extreme bull-market scenarios where the hedge underperforms, the short call is exercised, imposing a large liability on the collar holder.

For PnL versus Buy & Hold, across most percentiles PnL lags Buy & Hold. This persistent gap reflects the net cost of maintaining the collar as put premiums paid and call premium received do not fully cancel, and hedging friction further erodes returns. The 1st percentile of PnL (-0.36) is worse than Buy & Hold (-0.25), suggesting that in the most extreme downside scenarios the large negative call value can overwhelm the protective put payoff. At the 99th percentile the collar's PnL of 0.71 actually exceeds Buy & Hold's 0.60, reflecting that in the collar's best paths the delta hedge on the short call generates substantial profits. However this advantage is not representative of the broad middle of the distribution where Buy & Hold consistently dominates.

(6) In general, what do you notice about the subplots for putValue and callValue? What is the message regarding these two plots?

The put value histogram is dominated by an enormous spike at zero with a long, sparse right tail extending to about \$30. This is the shape for a long put payoff with a point mass at zero when the option expires worthless, and a positively skewed tail for the minority of paths in which the index falls below the strike.

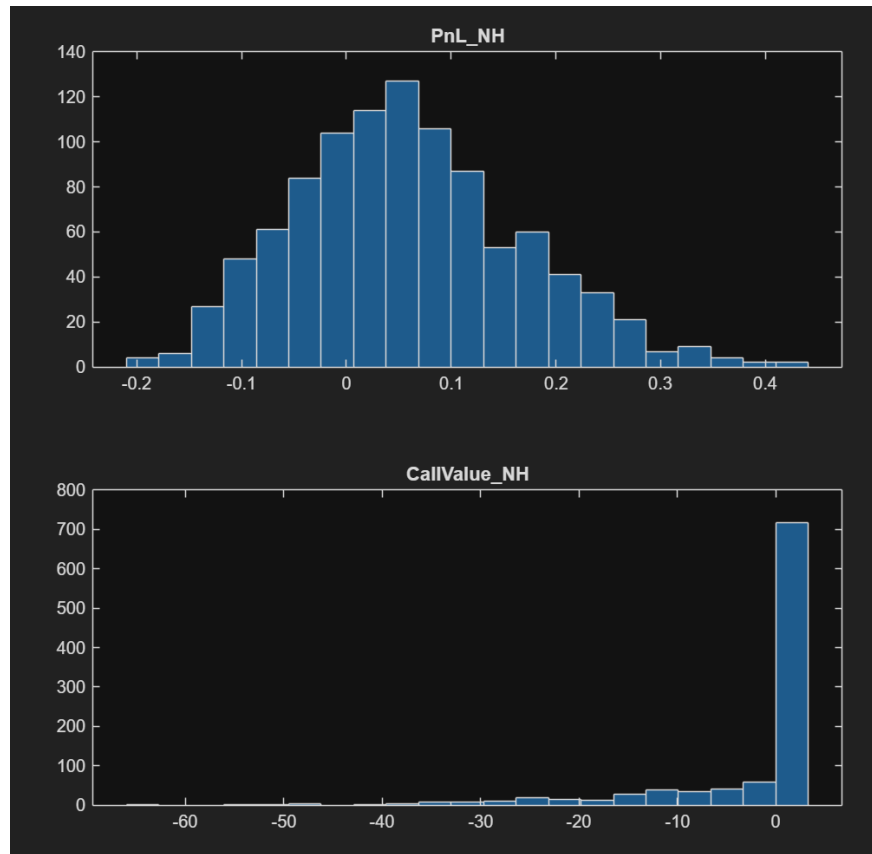
The call value histogram shows the opposite asymmetry. A dominant spike near zero, but with a left tail extending to around -\$45. The spike at zero corresponds to the many scenarios in which the index does not rally above the call strike and the short call expires worthless, costing nothing extra. The dangerous left tail represents strong bull-market runs where the short call is exercised against the collar holder at substantial cost.

Both distributions share the same structural feature, a concentrated mass at zero with a one-sided extreme tail. Most of the time options expire worthless and contribute nothing, but when they do

pay (or cost), the magnitude can be very large. The put's right tail delivers crash protection and the call's left tail represents the cost of that protection in bull markets.

(7) Look at the histogram for PnL unhedged (you might also look at the unhedged callValue_NH). What strikes you about these distributions and what could explain this behavior?

PnL is broader and more symmetric. The unhedged net P&L distribution (PnL) is noticeably more



symmetric and spread out than its hedged counterpart, ranging from roughly -0.50 to +0.50. It has an approximately bell-shaped appearance with a slight positive skew, consistent with the lognormal stock-price model's positive drift. This wider, more symmetric shape arises because without the hedge, the strategy's outcomes are driven almost entirely by the stock's terminal price rather than by path-dependent hedging costs.

For CallValue, the unhedged call value histogram is the most alarming of all the plots. Almost all observations crowd into a thin spike near zero (the premium retained when the call expires OTM, which is the most common outcome), but the distribution carries a catastrophic left tail extending beyond -\$60.

In this model, there is a nonzero and non-negligible probability of very large positive stock returns. When the index finishes far above the call strike, the short call is exercised deeply in-the-money, and the resulting loss can be many times the initial premium received. Without the stock position or a delta hedge to offset those gains, the collar holder bears the full brunt of those losses. The spike-and-catastrophic-tail shape of CallValue is precisely why people do not sell calls naked; delta hedging converts this distribution into the compact, near-zero bell curve seen in the hedged call subplot.

(8) In general, does it “pay” to invest in this collar strategy? (Compare PnL to “buy and hold”). Comparing Net P/L (Mean = 0.0752, Std = 0.2002) to the Buy & Hold distribution, the buy & hold strategy delivers higher returns across most of the distribution. Its median of 0.08 exceeds the collar’s 0.03, and at the 75th percentile Buy & Hold’s 0.23 surpasses the collar’s 0.19. At the 99th percentile the collar’s 0.71 exceeds Buy & Hold’s 0.60, though this reflects the independent nature of the percentile calculations rather than a genuine upside advantage. In a rising market, the short call caps portfolio gains, so the collar investor consistently sacrifices upside in bull scenarios. In this sense, the collar does not “pay” as it is a return-drag strategy across most of the distribution.

Where the collar earns its keep. Despite underperforming on expected return, the collar meaningfully compresses the left tail. The 1st percentile collar loss of -0.36 is worse than Buy & Hold’s -0.25, but the collar’s distribution is substantially tighter throughout the middle of the range (std 0.2002 vs. the wider Buy & Hold spread), giving investors more predictable outcomes. For investors with strong aversion to large drawdowns like pension funds, endowments, or individuals near retirement, this predictability and crash-protection has genuine economic value that a simple mean comparison obscures.

The collar strategy does not “pay” in the sense of maximizing expected return. A simple buy & hold dominates on that metric but it does “pay” as a risk-management tool. The long put provides a floor against severe market declines, while the short call (financed by its premium) partially offsets the cost of that protection.

Conclusion

Exercise Set 10 implements a dynamic delta-hedged collar strategy (long Stock A, long an ATM put, and short a 15% OTM call on the Index) simulated over 1,000 Monte Carlo paths, finding a mean Net P&L of 0.0752 with a standard deviation of 0.2002. The percentile analysis reveals that the put expires worthless in over half of all paths while the call’s hedged value swings between a catastrophic -\$45.08 at the 1st percentile and a large positive \$47.46 at the 99th, driven by the path-dependent delta hedge rather than the option payoff alone. The collar does not maximize expected return relative to a simple buy-and-hold strategy, but it earns its keep as a risk-management tool by compressing the middle of the return distribution and providing downside insurance in the rare but severe crash scenarios.

Appendix

Exercise Set 3: MatlabOLStutorial_26.m

```
%% Matlab OLS tutorial
%% 1. Clear and call data; define variables
clear;
clc;
cd ( '/Users/camja/ECON_4360 ' )
T = readtable( 'Econ_4360_DataSets_2026.xlsx', 'Sheet', 'Returns', 'Range',
    'A1:V384', 'VariableNamingRule', 'preserve' );
date = datetime(T.Date);
% Choose 4 indexes: USHY, USAGG, S&PLowVol, GSCI
X = [T.( "LF98TRUU Index" ), T.( "LBUSTRUU Index" ), T.( "SP5LVIT Index" ), T
    .( "SPGSCITR Index" ) ];
[N,k] = size(X);
%% 2. Statistics
X(1:10,:); % show first 10 rows, all 4 cols
mean(X);
std(X);
skewness(X(:,1:4)); % High Yield only
kurtosis(X(:,1:4));
% FYI how about annualizing these means/std?
(1+mean(X)).^12 -1;
sqrt(12)*std(X);
% Annualized returns to X
Xann = prod(1+X).^(12/N) - 1;
Xann = prod(1+[T.LF98TRUU, T.LBUSTRUU, T.SP5LVIT, T.SPGSCITR]).^(12/N)
    -1;

%% 3. Plotting
% Histograms
figure(1)
subplot(2,2,1); histfit(X(:,1)); title('High_Yield');
subplot(2,2,2); histfit(X(:,2)); title('US_Agg');
subplot(2,2,3); histfit(X(:,3)); title('S&PLowVol');
subplot(2,2,4); histfit(X(:,4)); title('Commodities');
% Time Series
figure(2)
subplot(2,1,1); plot(date,T.LF98TRUU); title('US_High_Yield'); grid on;
    recessionplot;
subplot(2,1,2); plot(date,T.SP5LVIT); title('Market_Risk_Premium');
    grid on; recessionplot;

% Cumulative value
Xcum = cumprod(1+X);
figure(3)
subplot(2,2,1); plot(date,Xcum(:,1)); recessionplot; title('US_High_
```



```

    Yield'); grid on;
subplot(2,2,2); plot(date,Xcum(:,2)); recessionplot; title('US_Agg');
    grid on;
subplot(2,2,3); plot(date,Xcum(:,3)); recessionplot; title('S&PLowVol')
    ; grid on;
subplot(2,2,4); plot(date,Xcum(:,4)); recessionplot; title('Commodities
    '); grid on;

%% 4. OLS Estimation using Matrix Algebra (BUILDING OFF OF REGRESSION
OVERVIEW.PPT)
%
% AQR factors: MKT, SMB, HML, BAB, UMD
% FundA (copied from BlackRock sheet) is appended as column G of the
AQR sheet.
% BAB begins Feb 1989 (Excel row 8); FundA ends Aug 2025 (Excel row
446).
% Range 'A1:G446' reads the header row plus all data through Aug 2025;
% trimming the first 6 data rows (Aug 1988 – Jan 1989) aligns the
table
% to Feb 1989, where every factor including BAB is available.
%
clear; clc;
T = readtable('Econ_4360_DataSets_2026.xlsx','Sheet','AQR','Range','A1:
G446','VariableNamingRule','preserve');
T = T(7:end,:); % drop Aug 1988 – Jan 1989 (BAB is NaN); data now Feb
1989 – Aug 2025

varNames = {'Alpha','MKT','SMB','HML','BAB','UMD'};
date = datetime(T.Date,'InputFormat','MM/dd/yyyy');
X = [T.MKT, T.SMB, T.HML, T.BAB, T.UMD];
[N,k] = size(X);
Y = T.FundA;
X = [ones(N,1), X]; % Add column of 1s for intercept (now N x 6)
b = inv(X'*X)*X'*Y; % parameter vector (6 x 1)
Yhat = X*b; % predicted values
ESS = sum((Yhat-Y).^2); % residual SS, df = N-k-1
RMSE = sqrt(ESS/(N-k-1)); % Root MSE
MSS = sum((Yhat-mean(Yhat)).^2); % model SS, df = k

% TSS = ESS + MSS (Total Sum of Squares = Error + Model)
Rsqr = MSS/(ESS+MSS); % R-squared

% Standard Errors
SSb = (ESS/(N-k-1))*inv(X'*X);
sb = sqrt(diag(SSb));

% t-stats H0: b(i) = 0
t = b./sb;

```

```

Results = table(b, sb, t, ...
    'VariableNames', { 'beta', 'stdError', 'tstat' }, ...
    'RowNames', { 'constant', 'MKT', 'SMB', 'HML', 'BAB', 'UMD' });

% Display results rounded to 3 decimal places
round(Results,3)

%% 5. Restricted regression ( $H_0: MKT = SMB = 0$ )
% Drop MKT (col 2) and SMB (col 3) from X; keep intercept, HML, BAB,
  UMD.

Xr      = [X(:,1), X(:,4:6)];          % intercept + HML, BAB, UMD
br      = inv(Xr'*Xr)*Xr'*Y;
Yhat_r  = Xr*br;
ESSr    = sum((Yhat_r-Y).^2);
F       = ((ESSr - ESS)/ESS)*(N-k-1)/2;

test    = 1 - fcd(f(F, 2, N-k-1));    % p-value (prob Type I error)
test

%% 6. 36-month Rolling regression

beta = zeros(N, k+1);                  % pre-allocate: N x 6

for t = 36:N
    Xr = X(t-35:t, :); Yr = Y(t-35:t);
    beta(t, :) = inv(Xr'*Xr)*Xr'*Yr;
end

figure(4)
subplot(3,2,1); plot(date(36:end), beta(36:end,2)); recessionplot;
    title('MKT'); grid on;
subplot(3,2,2); plot(date(36:end), beta(36:end,3)); recessionplot;
    title('SMB'); grid on;
subplot(3,2,3); plot(date(36:end), beta(36:end,4)); recessionplot;
    title('HML'); grid on;
subplot(3,2,4); plot(date(36:end), beta(36:end,5)); recessionplot;
    title('BAB'); grid on;
subplot(3,2,5); plot(date(36:end), beta(36:end,6)); recessionplot;
    title('UMD'); grid on;
subplot(3,2,6); plot(date(36:end), beta(36:end,1)); recessionplot;
    title('Alpha'); grid on;

%% 7. Optional: FITLM is more convenient when you use a table

tbl = table(T.MKT, T.SMB, T.HML, T.BAB, T.UMD, T.FundA, ...
    'VariableNames', { 'MKT', 'SMB', 'HML', 'BAB', 'UMD', 'Fund' });

```

```

LM = fitlm(tbl);
LM.Coefficients

%% 8. Return attribution

FundA = T.FundA;

% X already has a column of ones (intercept + 5 factors, k+1 = 6
    columns total)
[N,k] = size(X);          % k = 6 (intercept counts here)

% Pre-allocate storage
b_roll = zeros(N, k);      % rolling beta estimates    N x 6
A       = zeros(N, k+1);   % attribution matrix: idiosyncratic + k
    columns -> N x 7
% Column layout of A:
%   col 1 : idiosyncratic (residual) return
%   col 2 : alpha contribution (b_0 * 1)
%   col 3 : MKT contribution (b_MKT * MKT)
%   col 4 : SMB contribution (b_SMB * SMB)
%   col 5 : HML contribution (b_HML * HML)
%   col 6 : BAB contribution (b_BAB * BAB)
%   col 7 : UMD contribution (b_UMD * UMD)

for t = 36:N
    b_roll(t,:) = inv(X(t-35:t,:)'*X(t-35:t,:)) * X(t-35:t,:)'*FundA(
        t-35:t);
    A(t,:) = [FundA(t) - X(t,:)*b_roll(t,:)', b_roll(t,:).*X(t,:)]';
end

% Verify the attribution sums to FundA returns (should be ~0)
check = FundA - sum(A,2);

% Return Attribution plots
figure(5);
subplot(4,2,1); plot(date(36:end), A(36:end,1)); title('Idiosyn_Return');
    grid on;
subplot(4,2,2); plot(date(36:end), A(36:end,2)); title('Alpha');
    grid on;
subplot(4,2,3); plot(date(36:end), A(36:end,3)); title('MKT');
    grid on;
subplot(4,2,4); plot(date(36:end), A(36:end,4)); title('SMB');
    grid on;
subplot(4,2,5); plot(date(36:end), A(36:end,5)); title('HML');
    grid on;
subplot(4,2,6); plot(date(36:end), A(36:end,6)); title('BAB');
    grid on;

```

```

subplot(4,2,7); plot(date(36:end), A(36:end,7)); title('UMD');
                grid on;

%% 9. Optional – AQR annualized attribution

facRets = A; portRet = FundA;
for i = 47:N % TTM (rolling 12-month; start of rolling data is row
    36, so start calcs at 47)
    for j = 1:size(facRets,2)
        retAnn(i,j) = sum(facRets(i-11:i,j)) + ...
            ( (prod(1+portRet(i-11:i))-1) - sum(sum(facRets(i-11:i,:)))
            ) * ...
            abs(sum(facRets(i-11:i,j))) / sum(abs(ones(1,12)*facRets(i
            -11:i,:)));
    end
    compound(i) = prod(1+portRet(i-11:i)) - 1;
end

checkAnnual = compound' - sum(retAnn,2); % Check the math (should be
    ~0)

figure(6);
subplot(4,2,1); plot(date(47:end), retAnn(47:end,1)); title('Idiosyn□
    Return'); grid on;
subplot(4,2,2); plot(date(47:end), retAnn(47:end,2)); title('Alpha');
                grid on;
subplot(4,2,3); plot(date(47:end), retAnn(47:end,3)); title('MKT');
                grid on;
subplot(4,2,4); plot(date(47:end), retAnn(47:end,4)); title('SMB');
                grid on;
subplot(4,2,5); plot(date(47:end), retAnn(47:end,5)); title('HML');
                grid on;
subplot(4,2,6); plot(date(47:end), retAnn(47:end,6)); title('BAB');
                grid on;
subplot(4,2,7); plot(date(47:end), retAnn(47:end,7)); title('UMD');
                grid on;

% Heat maps
% varNames has 6 entries: Alpha + 5 AQR factors (columns 2–7 of retAnn
    / b_roll)
dates = datestr(date(47:end), 'mm/yyyy');

% Attribution heat map (annualized factor contributions, cols 2:7 =
    alpha + 5 factors)
curTbl = array2table(retAnn(47:end, 2:end), ...
    'VariableNames', varNames, ...
    'RowNames', cellstr(dates));
figure;

```

```

heatmapfig1 = lmheatmap(curTbl, 'Fund');

% Rolling-beta heat map (cols 1:6 of b_roll = intercept + 5 factors)
curTbl = array2table(b_roll(47:end,:), ...
    'VariableNames', varNames, ...
    'RowNames',      cellstr(dates));
figure;
heatmapfig2 = lmheatmap(curTbl, 'Fund');

```

Exercise Set 5.m

```

%% Step 1: Load data, build X and benchmark Xb
clear; clc;

% Read the Index sheet (row 2 = Bloomberg tickers as headers)
T = readtable('Econ_4360_DataSets_2026.xlsx', ...
    'Sheet', 'Index', ...
    'VariableNamingRule', 'preserve');

% Extract the date column
date = T{:,1}; % datetime array

% — Portfolio 1 indices (from Sample Ports sheet) —
% R2000, MSCI ACWI, DJUS REIT TR, 7–10 YR TREAS, US AGG, S&P GSCI
% Bloomberg tickers:
% RTY Index, MXWD Index, DWRIFT Index, LT09TRUU Index, LBUSTRUU Index,
% SPGSCITR Index
% Build price matrix (each column = one index, in levels)
P_R2000 = T{:, 'RTY_Index'};
P_MXWD = T{:, 'MXWD_Index'};
P_REIT = T{:, 'DWRIFT_Index'};
P_TREAS710 = T{:, 'LT09TRUU_Index'};
P_USAGG = T{:, 'LBUSTRUU_Index'};
P_GSCI = T{:, 'SPGSCITR_Index'};
% Convert price levels to monthly log returns
prices = [P_R2000, P_MXWD, P_REIT, P_TREAS710, P_USAGG, P_GSCI];
X = diff(log(prices)); % T-1 x 6 matrix of log returns
date = date(2:end); % align dates
% — Benchmark: 40% LEGATRUH (Global Agg) / 60% MXWD —
P_LEGATRUH = T{:, 'LEGATRUH_Index'};
P_MXWD2 = T{:, 'MXWD_Index'};
bench_prices = [P_LEGATRUH, P_MXWD2];
bench_ret = diff(log(bench_prices));
Xb = bench_ret * [0.4; 0.6]; % benchmark return series (scalar per
    period)

%% Step 2: Estimate covariance, mean returns, std devs
[T_obs, N] = size(X);

V = cov(X); % 6x6 covariance matrix (monthly)
mu = (prod(1 + X) .^ (1/T_obs) - 1)'; % 6x1 monthly geometric mean
    returns
s = std(X)'; % 6x1 monthly standard
    deviations

```

```

disp('Covariance_matrix_V:'); disp(V)
disp('Monthly_geometric_mean_returns_mu:'); disp(mu)
disp('Monthly_std_devs_s:'); disp(s)

%% Step 3: Maximum Sharpe ratio (tangency) portfolio
w_raw = V \ mu; % equivalent to inv(V)*mu

% Normalize so weights sum to 1
w = w_raw / sum(w_raw);

% Display
index_names = {'USHY', 'US_Agg', 'SP_Low_Vol', 'SP_GSCI', 'R2000', 'EPRA_NAREIT'};
disp('Tangency_portfolio_weights:');
disp(table(index_names', w, 'VariableNames', {'Index', 'Weight'}))

%% Step 4: Efficient frontier using portopt
% portopt(mu', V) returns [PortRisk, PortReturn, PortWts]
% mu must be a row vector for portopt

NumPorts = 100;
[PortRisk, PortReturn, PortWts] = portopt(mu', V, NumPorts);

% — Plot frontier —
figure;
plot(PortRisk, PortReturn, 'b-', 'LineWidth', 2);
xlabel('Portfolio_Risk_(Monthly_Std_Dev)');
ylabel('Portfolio_Return_(Monthly)');
title('Efficient_Frontier');
grid on;

% Minimum variance portfolio = first point (leftmost)
mvp_risk = PortRisk(1);
mvp_return = PortReturn(1);

% Tangency portfolio = closest point on frontier to our w
% Re-compute tangency risk/return from w
tang_return = w' * mu;
tang_risk = sqrt(w' * V * w);

% Mark both on plot
hold on;
plot(mvp_risk, mvp_return, 'ro', 'MarkerSize', 10, 'DisplayName', 'Min_Variance');
plot(tang_risk, tang_return, 'gs', 'MarkerSize', 10, 'DisplayName', 'Tangency');
legend('Frontier', 'Min_Variance', 'Tangency');

```

```

%% Step 5: Portfolio return time series, annualized stats
N = length(X);           % number of monthly observations

R = X * w;               % Tx1 portfolio monthly return series
Rb = Xb;                 % benchmark already computed in Step 1

% Annualized geometric return
ann_ret = prod(1 + R) ^ (12/N) - 1;
ann_retb = prod(1 + Rb) ^ (12/N) - 1;

% Annualized volatility
ann_vol = sqrt(12) * std(R);
ann_volb = sqrt(12) * std(Rb);

fprintf('\n—— Portfolio ——\n')
fprintf('Annualized Return: %0.2f%%\n', ann_ret * 100)
fprintf('Annualized Volatility: %0.2f%%\n', ann_vol * 100)

fprintf('\n—— Benchmark ——\n')
fprintf('Annualized Return: %0.2f%%\n', ann_retb * 100)
fprintf('Annualized Volatility: %0.2f%%\n', ann_volb * 100)

%% Step 6: $1 cumulative growth plot
Rgrowth = cumprod(1 + R);
Rbgrowth = cumprod(1 + Rb);

figure;
plot(date, [Rgrowth, Rbgrowth], 'LineWidth', 1.5);
title('Growth of $1 Investment');
xlabel('Date'); ylabel('Portfolio Value ($)');
grid on;
recessionplot;
legend('Portfolio', 'Benchmark');

%% Step 7: Risk attribution – MCR and ATTR
port_vol = sqrt(w' * V * w); % portfolio volatility (monthly)

% Marginal Contribution to Risk: MCR_i = (V*w)_i / port_vol
MCR = (V * w) / port_vol; % 6x1

% Attribution to Risk: ATTR_i = w_i * MCR_i
ATTR = w .* MCR; % 6x1 (should sum to port_vol)

% Annualize for display
ann_port_vol = sqrt(12) * port_vol;

fprintf('\n—— Risk Attribution (annualized) ——\n')
T_risk = table(index_names', w, MCR*sqrt(12), ATTR*sqrt(12), ...

```



```

    'VariableNames', {'Index', 'Weight', 'MCR_ann', 'ATTR_ann'});
disp(T_risk)
fprintf('Portfolio Annualized Volatility: %.4f\n', ann_port_vol)
fprintf('Sum of ATTR (annualized): %.4f\n', sum(ATTR)*sqrt(12))

%% Step 8: Rolling 36-month factor betas for R and Rb

% Load BlackRock factors
TB = readtable('Econ_4360_DataSets_2026.xlsx', ...
    'Sheet', 'BlackRock', ...
    'VariableNamingRule', 'preserve');

factor_date = TB{:,1};
factor_names = TB.Properties.VariableNames(2:7); % 6 factor names
Factors = TB{:, 2:7}; % Tx6

% Align dates: find overlapping date range between R and Factors
[common_dates, iR, iF] = intersect(date, factor_date);
R_aligned = R(iR);
Rb_aligned = Rb(iR);
F_aligned = Factors(iF, :);
N_aligned = length(common_dates);

win = 36; % 36-month rolling window
n_rolls = N_aligned - win + 1;

beta_R = NaN(n_rolls, 7); % 6 betas + intercept
beta_Rb = NaN(n_rolls, 7);

for t = 1:n_rolls
    idx = t : t + win - 1;
    F_win = [ones(win,1), F_aligned(idx,:)]; % add constant

    % OLS: beta = (F'F)^-1 * F'y
    beta_R(t,:) = (F_win \ R_aligned(idx))';
    beta_Rb(t,:) = (F_win \ Rb_aligned(idx))';
end

roll_dates = common_dates(win : end);

% — Plot betas for Portfolio —
figure;
plot(roll_dates, beta_R(:, 2:end), 'LineWidth', 1.2); % skip
    intercept col
title('Rolling_36M_Factor_Betas Portfolio');
xlabel('Date'); ylabel('Beta');
legend(factor_names, 'Location', 'best'); grid on;

```

```

% — Plot betas for Benchmark —
figure;
plot(roll_dates, beta_Rb(:, 2:end), 'LineWidth', 1.2);
title('Rolling_36M_Factor_Betas_Benchmark');
xlabel('Date'); ylabel('Beta');
legend(factor_names, 'Location', 'best'); grid on;

%% Step 9: Rolling returns attribution for R and Rb

% Attribution for each window = beta_i * factor_mean_i over that window
attr_R = NaN(n_rolls, 6); % 6 factors (exclude intercept)
attr_Rb = NaN(n_rolls, 6);

for t = 1:n_rolls
    idx = t : t + win - 1;
    F_mean = mean(F_aligned(idx,:), 1); % 1x6 mean factor returns in
        window

    attr_R(t,:) = beta_R(t, 2:end) .* F_mean; % element-wise
    attr_Rb(t,:) = beta_Rb(t, 2:end) .* F_mean;
end

% — Plot attributions Portfolio —
figure;
area(roll_dates, attr_R);
title('Rolling_36M>Returns_Attribution_Portfolio');
xlabel('Date'); ylabel('Attribution(monthly)');
legend(factor_names, 'Location', 'best'); grid on;

% — Plot attributions Benchmark —
figure;
area(roll_dates, attr_Rb);
title('Rolling_36M>Returns_Attribution_Benchmark');
xlabel('Date'); ylabel('Attribution(monthly)');
legend(factor_names, 'Location', 'best'); grid on;

%% Compute average attribution percentages over 2010–2020

% Find row indices corresponding to 2010–2020 in roll_dates
idx_period = year(roll_dates) >= 2010 & year(roll_dates) <= 2020;

% Average attribution per factor over that period
attr_R_period = mean(attr_R(idx_period, :), 1); % 1x6 vector

% Total average return attributed to all factors combined
total_attr = sum(abs(attr_R_period));

% Percentage each factor contributes

```

```

attr_pct = (attr_R_period / total_attr) * 100;

% Display
T_attr = table(factor_names', attr_R_period', attr_pct', ...
    'VariableNames', {'Factor', 'Avg_Attribution', 'Pct_of_Total'});
disp('——ReturnsAttribution(2010–2020)——')
disp(T_attr)

%% Step 10: Heat maps of returns attribution

% —— Portfolio heat map ——
figure;
imagesc(attr_R');
colorbar;
colormap(cool);
title('ReturnsAttributionHeatMapPortfolio');
xlabel('RollingWindow');
ylabel('Factor');
yticks(1:6); yticklabels(factor_names);
% Optional: label x-axis with years
yr_ticks = 1 : floor(n_rolls/10) : n_rolls;
xticks(yr_ticks);
xticklabels(datestr(roll_dates(yr_ticks), 'yyyy'));
xtickangle(45);

% —— Benchmark heat map ——
figure;
imagesc(attr_Rb');
colorbar;
colormap(cool);
title('ReturnsAttributionHeatMapBenchmark');
xlabel('RollingWindow');
ylabel('Factor');
yticks(1:6); yticklabels(factor_names);
xticks(yr_ticks);
xticklabels(datestr(roll_dates(yr_ticks), 'yyyy'));
xtickangle(45);

%% Benchmark attribution 2010–2020
attr_Rb_period = mean(attr_Rb(idx_period, :), 1);
total_attr_b = sum(abs(attr_Rb_period));
attr_pct_b = (attr_Rb_period / total_attr_b) * 100;

T_attr_b = table(factor_names', attr_Rb_period', attr_pct_b', ...
    'VariableNames', {'Factor', 'Avg_Attribution_Benchmark', 'Pct_of_Total'});
disp('——BenchmarkAttribution(2010–2020)——')
disp(T_attr_b)

```

```

%% Side by side comparison
T_compare = table(factor_names', attr_pct', attr_pct_b', attr_pct' -
    attr_pct_b', ...
    'VariableNames', {'Factor', 'Portfolio_Pct', 'Benchmark_Pct', '
        Difference'});
disp('—— Portfolio vs Benchmark Attribution ——')
disp(T_compare)

```

Exercise Set 6: OptPort_completed-1.m

```
%% 1. Import Indexes, Create data set Z, Vcov, mean_returns
clear;
clc;

% cd('/Users/stevenpeterson/Desktop/ECON 4360')

T = readtable('Econ_4360_DataSets_2026_(1).xlsx', ...
    'Sheet','Returns', ...
    'Range','A1:V384');

date = datetime(T.Date);

% Choose 6 indexes: USHY, USAGG, S&PLowVol, GSCI, R2000, EPRA_NAREIT
X = [T.LF98TRUU, T.LBSTRUU, T.SP5LVIT, T.SPGSCITR, T.RTY, T.RUGL];

[N,k] = size(X);

% Benchmark portfolio (opportunity cost portfolio)
Xb = [T.LEGATRUH, T.MXWD] * [0.4; 0.6];

Vcov = cov(X);
[s,C] = cov2corr(Vcov);

mean_return = prod(1+X).^(1/N) - 1;

%% 2. Portfolio Optimization (full sample)

nAssets = numel(mean_return);
r = 0.0055;

Aeq = ones(1,nAssets);
beq = 1;

Aineq = -mean_return;
bineq = -r;

lb = zeros(nAssets,1);
ub = ones(nAssets,1);

c = zeros(nAssets,1);

options = optimset('Algorithm','interior-point-convex', ...
    'Display','off', ...
    'TolFun',1e-10);
```

```

wmv = quadprog(Vcov,c,[],[],Aeq,beq,lb,ub,[],options);
wSharpe = quadprog(Vcov,c,Aineq,bineq,Aeq,beq,lb,ub,[],options);

wRP = OptRP(Vcov);

%% 3. Performance (Returns and Risks)

r_mv = mean_return * wmv;
r_Sharpe = mean_return * wSharpe;
r_RP = mean_return * wRP;

rmv_yoy = (1+r_mv)^12 - 1;
rSharpe_yoy = (1+r_Sharpe)^12 - 1;
rRPyoy = (1+r_RP)^12 - 1;

sigma_mv = sqrt(wmv' * Vcov * wmv);
sigma_Sharpe = sqrt(wSharpe' * Vcov * wSharpe);
sigma_RP = sqrt(wRP' * Vcov * wRP);

corr_mv_Sharpe = (wmv' * Vcov * wSharpe)/(sigma_mv*sigma_Sharpe);
corr_mv_RP = (wmv' * Vcov * wRP)/(sigma_mv*sigma_RP);
corr_Sharpe_RP = (wSharpe' * Vcov * wRP)/(sigma_Sharpe*sigma_RP);

ExpCorr = [1, corr_mv_Sharpe, corr_mv_RP;
           corr_mv_Sharpe, 1, corr_Sharpe_RP;
           corr_mv_RP, corr_Sharpe_RP, 1];

ExpSigma = diag([sigma_mv, sigma_Sharpe, sigma_RP]);
ExpCov = ExpSigma * ExpCorr * ExpSigma;

```

Exercise Set 7.m

```
%% Exercise Set 7 – Retirement Annuity (Complete Solution)
clear; clc;

%% Parameter settings
T = 30; % career length in years
N = 10000; % number of Monte Carlo simulations
sigma = 0.10; % asset return volatility
r = 0.07; % mean annual return
g = 0.04; % salary growth rate
s = 0.10; % annual savings rate
P = 65000; % starting salary

%% Preallocate arrays
Y = zeros(T,N); % Salary matrix [years x simulations]
S = zeros(T,N); % Savings (future value) matrix
W = zeros(T,N); % Wealth matrix

Y(1,:) = P; % First year salary for all N paths
S(1,:) = s*P*(1+r)^(T-1);
W(1,:) = S(1,1);

%% Monte Carlo simulation
for n = 1:N
    R = r + sigma*randn(T,1);

    for t = 2:T
        Y(t,n) = (1+g)*Y(t-1,n);
        S(t,n) = (s*Y(t,n)) * prod(1 + R(t:T));
        W(t,n) = sum(S(1:t,n));
    end
end

%% Q1: Terminal wealth distribution
figure(1);
histogram(W(end,:), 60);
title('End-of-Career Wealth Distribution (N=10,000 simulations)');
xlabel('Terminal Wealth ($)');
ylabel('Frequency');
xline(median(W(end,:)), 'r—', 'LineWidth', 2, 'Label', 'Median');

med_W = median(W(end,:));
std_W = std(W(end,:));

fprintf('\n——Q1: Terminal Wealth——\n');
fprintf('Median: %.2f\n', med_W);
```

```

fprintf('Std_Dev:_%$.2f\n', std_W);

%% Q2: Percentiles and IQR
pcts = prctile(W(end,:), [1 5 25 50 75 95 99]);

fprintf('\n——Q2: Percentiles——\n');
fprintf('1st_pct:_%$.0f\n', pcts(1));
fprintf('5th_pct:_%$.0f\n', pcts(2));
fprintf('25th_pct:_%$.0f\n', pcts(3));
fprintf('50th_pct:_%$.0f\n', pcts(4));
fprintf('75th_pct:_%$.0f\n', pcts(5));
fprintf('95th_pct:_%$.0f\n', pcts(6));
fprintf('99th_pct:_%$.0f\n', pcts(7));
fprintf('IQR:_%$.0f\n', pcts(5) - pcts(3));

%% Q3: Downside risk
P_thresh = 946586;
prob_below = mean(W(end,:) < P_thresh);

fprintf('\n——Q3: Downside Risk——\n');
fprintf('Prob<_%$.0f=_%$.1f%%\n', P_thresh, 100*prob_below);

%% Q4-5: Retirement annuity
T_ret = 30;
PVA = (1 - (1+r)^(-T_ret)) / r;
A = med_W / PVA;

fprintf('\n——Q4-5: Retirement Annuity——\n');
fprintf('Median_wealth:_%$.2f\n', med_W);
fprintf('PVA:_%$.4f\n', PVA);
fprintf('Annual_annuity:_%$.2f\n', A);
fprintf('Monthly_annuity:_%$.2f\n', A/12);

%% Q6: Medical expense shock simulation
M = 50000;
p_shock = 0.05;
N_ret = 10000;

W_ret = zeros(T_ret+1, N_ret);
W_ret(1,:) = med_W;

for n = 1:N_ret
    R_ret = r + sigma*randn(T_ret,1);

    for t = 1:T_ret
        shock = M * (rand < p_shock);
        W_ret(t+1,n) = W_ret(t,n)*(1+R_ret(t)) - A - shock;
    end
end

```



```

        if W_ret(t+1,n) < 0
            W_ret(t+1:end,n) = 0;
            break;
        end
    end
end

prob_ruin = mean(W_ret(end,:) <= 0);

fprintf( '\n——Q6: Medical Shock——\n' );
fprintf( 'Shock: $%.0f | Prob: %.1f%%\n', M, 100*p_shock );
fprintf( 'Ruin probability: %.1f%%\n', 100*prob_ruin );

figure(2);
histogram(W_ret(end,:), 60);
title( 'Retirement Wealth with Medical Shocks' );
xlabel( 'Terminal Wealth ($)' );
ylabel( 'Frequency' );

%% Sanity check
W_det = s*P * ((1+r)^T - (1+g)^T) / (r - g);

fprintf( '\n——Sanity Check——\n' );
fprintf( 'Deterministic W_T: $%.2f\n', W_det );

```

Exercise Set 9: portInsuranceHedge.m

```

%% 1. Parameters
clear; clc;
format bank
Kp = 90; s_A = 0.142; s_I = 0.15; rho = 0.90; r = 0.04;
Exp_returns = [0.093, 0.06]; % [max Sharpe portfolio E[r], Index E[r]]
C = [1 rho; rho 1];
S = diag([s_A, s_I]);
V = S*C*S;
returns = Exp_returns;
Week = [0:1:52]';
Time = (repmat(52, length(Week), 1) - Week) / 52;
Time(end) = (Week(end) + 0.5 - 52) / 52;
% End Parameters

%% 2. Simulate Stock A and Index (Stock A = your maxSharpe portfolio
for n = 1:1000
M = [100 100; zeros(52, 2)]; % Stock A, Index
for t = 2:length(Time)
    M(t, :) = M(t-1, :). * exp(returns/52 + randn(1, 2) * chol(V) / sqrt(52));
end

d1p = (log(M(:, 2) / Kp) + repmat(r, length(Time), 1) + 0.5 * repmat(s_I^2, length(
    Time), 1) * Time) ./ (s_I * sqrt(Time)));
d2p = d1p - s_I * sqrt(Time);

%% 3. Price option and put delta

put = repmat(Kp, length(Time), 1) .* exp(-r * Time) .* normcdf(-d2p) - M(:, 2) .*
    normcdf(-d1p);
delta = -normcdf(-d1p);

%% 4. Delta Hedge Accounting

X = [delta(1), -100 * delta(1), -r * 100 * delta(1) / 52; zeros(52, 3)];
for t = 2:length(Time)
    X(t, 1) = delta(t) - delta(t-1);
    X(t, 2) = X(t-1, 2) - X(t, 1) * M(t, 2) + X(t-1, 3); % Cash bal
    X(t, 3) = X(t, 2) * r / 52; % interest
end
DeltaHedgeAcct = table(Week, Time, M(:, 2), M(:, 1), d1p, d2p, put, delta, X(:, 2),
    X(:, 3), ...
    'VariableNames', {'Week', 'Time', 'Index', 'StockA', 'd1pI', 'd2p', '
    Put', 'Delta', 'Cash', 'Interest'});

%% 5. Portfolio Value with Hedge
putValueHedged(n) = DeltaHedgeAcct.Put(1) + DeltaHedgeAcct.Index(end) -
    Kp;

```

```

putUnhedged(n) = DeltaHedgeAcct.Put(1) - (Kp>DeltaHedgeAcct.Index(end)
    )*(Kp-DeltaHedgeAcct.Index(end));

end

%% 6. Export Results to Excel
% Single-path hedge accounting table (last simulation run)
writetable(DeltaHedgeAcct, 'DeltaHedgeAcct.csv');

% Summary statistics across all 1000 paths
SummaryStats = table(...
    {'Hedged'; 'Unhedged'}, ...
    [mean(putValueHedged); mean(putUnhedged)], ...
    [std(putValueHedged); std(putUnhedged)], ...
    [min(putValueHedged); min(putUnhedged)], ...
    [max(putValueHedged); max(putUnhedged)], ...
    [mean(putValueHedged<0); mean(putUnhedged<0)], ...
    'VariableNames', {'Strategy', 'Mean', 'StdDev', 'Min', 'Max', 'ProbLoss'
    });
disp(SummaryStats);
writetable(SummaryStats, 'SimulationSummary.csv');

% All 1000 payoffs for both strategies
AllPayoffs = table((1:1000)', putValueHedged', putUnhedged', ...
    'VariableNames', {'Simulation', 'HedgedPL', 'UnhedgedPL'});
writetable(AllPayoffs, 'AllSimulationPayoffs.csv');

%% 6. Export Results to Excel
outDir = 'C:\Users\camja\Downloads\';

% Single-path hedge accounting table (last simulation run)
writetable(DeltaHedgeAcct, [outDir 'DeltaHedgeAcct.csv']);

% Summary statistics across all 1000 paths
SummaryStats = table(...
    {'Hedged'; 'Unhedged'}, ...
    [mean(putValueHedged); mean(putUnhedged)], ...
    [std(putValueHedged); std(putUnhedged)], ...
    [min(putValueHedged); min(putUnhedged)], ...
    [max(putValueHedged); max(putUnhedged)], ...
    [mean(putValueHedged<0); mean(putUnhedged<0)], ...
    'VariableNames', {'Strategy', 'Mean', 'StdDev', 'Min', 'Max', 'ProbLoss'
    });
disp(SummaryStats);
writetable(SummaryStats, [outDir 'SimulationSummary.csv']);

% All 1000 payoffs for both strategies
AllPayoffs = table((1:1000)', putValueHedged', putUnhedged', ...

```

```
    'VariableNames', {'Simulation', 'HedgedPL', 'UnhedgedPL'});  
writetable(AllPayoffs, [outDir 'AllSimulationPayoffs.csv']);
```

Exercise Set 10: deltaHedgeTemplate.m

```

clear; clc;
%% 1. Parameters
format bank
Kp = 100; Kc = 115;
%Replace s_A, rho, returns with your portfolio moments
s_A = 0.192; s_I = 0.18; rho = 0.911; r = 0.02;
returns = [0.093, 0.08];
C = [1 rho; rho 1];
S = [s_I, 0; 0, s_A];
V = S * C * S;
Week = [0:1:52]';
Time = ( repmat(52,length(Week),1)-Week)/52;
Time(end) = (Week(end)+0.5-52)/52; % non-zero ending Time
N = 1000;
% End Parameters

%% 2. Simulate Spot Prices for Portfolio and Index beginning with 100
for n = 1:N

M = [100 100; zeros(52,2)]; % Stock A, Index
for t = 2:length(Time)
    M(t,:) = M(t-1,:).*exp((returns - 0.5*[s_A^2,s_I^2])/52 + flipplr(
        randn(1,2)*chol(V,'lower')))/sqrt(52));
end
% Now do the entire columns for the following:
d1c = (log(M(:,2)/Kc) + (r + 0.5*s_I^2)*Time) ./ (s_I*sqrt(Time));
d2c = d1c - s_I*sqrt(Time);
d1p = (log(M(:,2)/Kp) + (r + 0.5*s_I^2)*Time) ./ (s_I*sqrt(Time));
d2p = d1p - s_I*sqrt(Time);

%% 3. Price options and Compute delta (do the entire column(s))

put = Kp.*exp(-r.*Time).*normcdf(-d2p) - M(:,2).*normcdf(-d1p);
call = M(:,2).*normcdf(d1c) - Kc.*exp(-r.*Time).*normcdf(d2c);
delta = normcdf(d1c);

%% 4. Delta Hedge Accounting (X is cols Q, R, T in spreadsheet)

X = [delta(1),100*delta(1),r*100*delta(1)/52;zeros(52,3)];
for t = 2:length(Time)
    X(t,1) = delta(t) - delta(t-1);
    X(t,2) = X(t-1,2) + X(t,1)*M(t,2) + X(t-1,3);
    X(t,3) = r*X(t,2)/52;
end

%% 5. DeltaHedgeAcct

```

```

% Construct a table replicating cell range G1:T54 (with a header).
%DeltaHedgeAcct = table(Week,Time,M(:,2),M(:,1),d1c,d2c,d1p,d2p,put,
    call,delta,X(:,1),X(:,2),X(:,3),...
%     'VariableNames',{ 'Week','Time','Index','StockA','d1c','d2c','d1p
    ','d2p','Put','Call','Delta','Shares','Cost','Interest'});

%% 6. Portfolio Value with Hedge (Profit Accounting rows 57-61 etc.)
% Notes: putValue, callValue,callValue_NH, costHedge in $$
% Notes: PnL, PnL_NH in % of beginning $100
putValue      = max(Kp - M(end,2), 0);
callValue     = call(1) + (M(end,2)>Kc)*Kc - X(end,2);
callValue_NH  = call(1) - max(M(end,2)-Kc, 0);
PnL           = (M(end,1)-100 + putValue - put(1) + callValue) / 100;
PnL_NH        = (M(end,1)-100 + putValue - put(1) + callValue_NH) / 100;
DynamicHedge(n,:) = [putValue,callValue,PnL,(M(end,1)-M(1,1))/100,
    PnL_NH, callValue_NH];
costHedge(n) = (M(end,2)>Kc)*Kc - X(end,2);

end

%% Loop finished. Now summarize what you've analyzed
figure(1)
subplot(2,2,1); histogram(DynamicHedge(:,1),21); title('put_value')
subplot(2,2,2); histogram(DynamicHedge(:,2),21); title('call_value')
subplot(2,2,3); histogram(DynamicHedge(:,3),21); title('PnL')
subplot(2,2,4); histogram(DynamicHedge(:,4),21); title('Buy&Hold')

% Question 3: Net P/L histogram with mean and std
figure(3)
histogram(DynamicHedge(:,3),21);
title('Net_P/L(Hedged)_N=1000');
xlabel('Net_P/L(fraction of $100)'); ylabel('Frequency');
fprintf('Net_P/L_Mean=%.4fStd=%.4f\n', mean(DynamicHedge(:,3)),
    , std(DynamicHedge(:,3)));

% Percentiles for DynamicHedge(:,1:4)
Centile = prctile(DynamicHedge(:,1:4),[1 25 50 75 99]);
PercentileTable = table({'1st','25th','50th','75th','99th'}, ...
    Centile(:,1),Centile(:,2),Centile(:,3),Centile(:,4),...
    'VariableNames',{'Percentile','PutValue','callValue','PnL','
    Buy_and_Hold'})

figure(2)
subplot(2,1,1); histogram(DynamicHedge(:,5),21); title('PnL_NH')
subplot(2,1,2); histogram(DynamicHedge(:,6),21); title('CallValue_NH')

```